

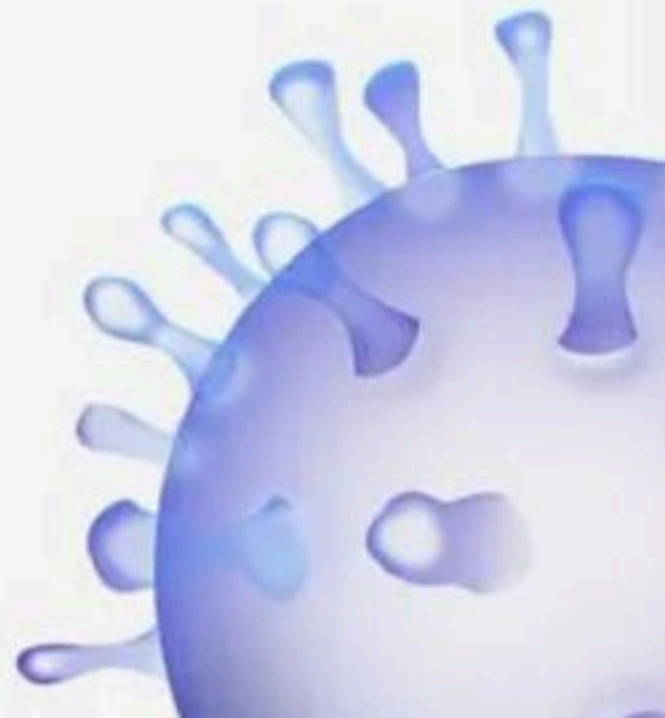
200 MCQs Biology - Model 1

NCERT Nichod : Biology Practice Series NEET 2025



200 MCQs

Model – 1 Test



BIOLOGYLIVE

1. Select a genus not related to family Solanaceae

- (1) Solanum
- (2) Mangifera
- (3) Petunia
- (4) Datura



2. Assertion : Systematics is a branch of biology that deals with classification of organisms

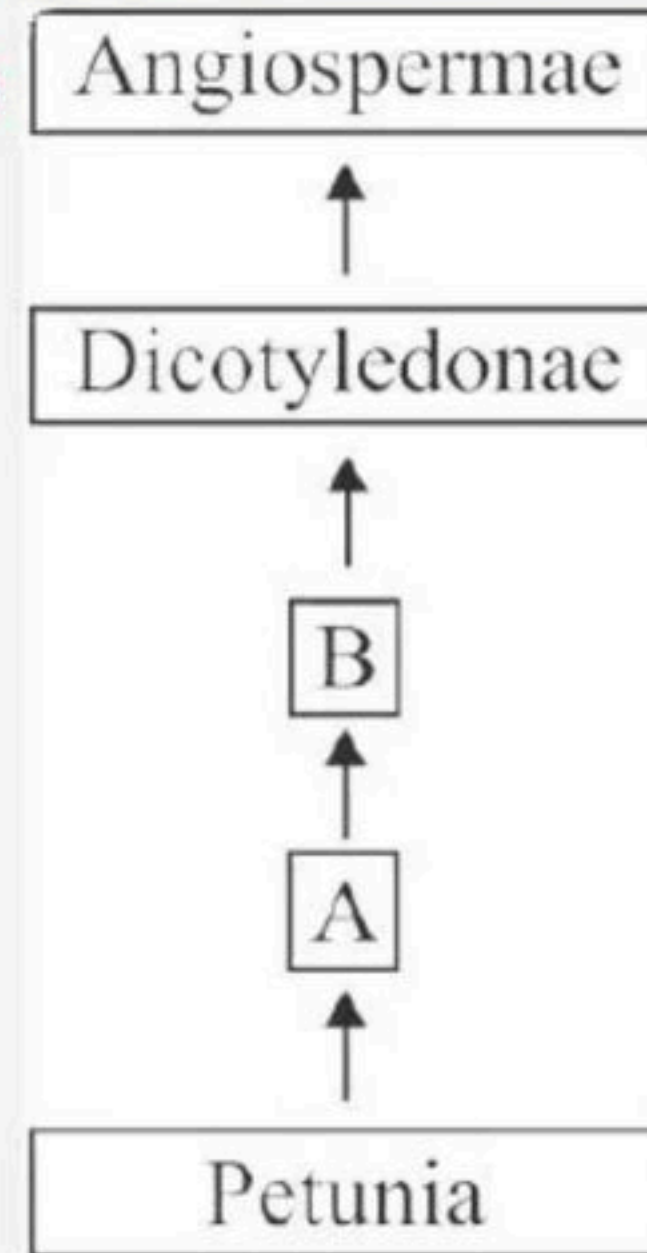
Reason : Aim of classification is to group the organisms in orderly manner

- (1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.



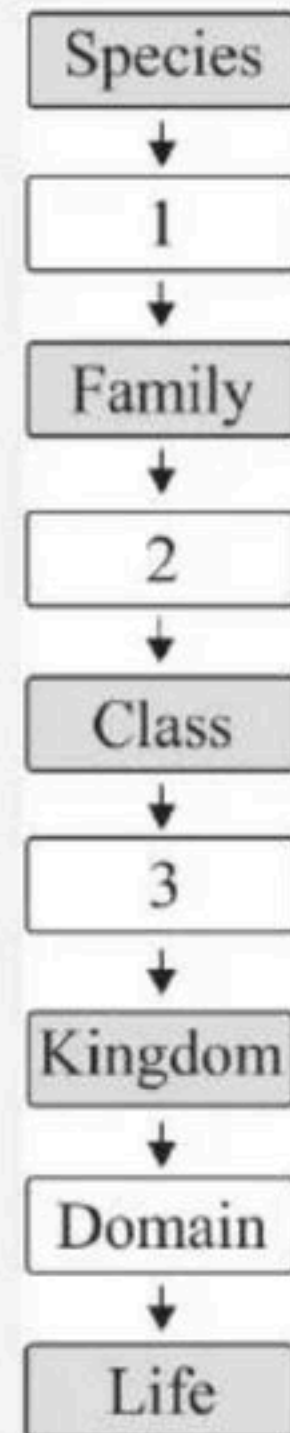
3. Recognise the following flow diagram and find the correct option according to taxonomic hierarchy.

- (1) A - Solanaceae, B - Polymoniales
- (2) A - Polymoniales, B - Convolvulaceae
- (3) A - Polymoniales, B - Solanaceae
- (4) A - Sapindales, B - Anacardiaceae



4. Choose the correct option for the following figure.

	1	2	3
(1)	Genus	Order	Phylum
(2)	Phylum	Order	Genus
(3)	Order	Phylum	Genus
(4)	Order	Genus	Phylum



5. Match the following columns and select the correct choice

Column I		Column II	
A	Family	P	melongena
B	Kingdom	Q	Polymoniales
C	Order	R	Datura
D	Species	S	Plantae
E	Genus	T	Solanaceae

(1) A - T, B - S, C - Q, D - P, E - R

(2) A - T, B - Q, C - S, D - P, E - R

(3) A - P, B - Q, C - R, D - S, E - T

(4) A - P, B - Q, C - S, D - R, E - T



6. Match the following and choose the correct option:

Column I		Column II	
A	Family	P	tuberosum
B	Kingdom	Q	Polymoniales
C	Order	R	Solanum
D	Species	S	Plantae
E	Genus	T	Solanaceae

(1) P-D, Q-C, R-E, S-B, T-A

(2) P-E, Q-D, R-B, S-A, T-C

(3) P-D, Q-E, R-B, S-A, T-C

(4) P-E, Q-C, R-B, S-A, T-D



7. Means of consciousness

I. All organisms are aware of their surrounding

II. All organisms are stimulated by their surrounding

III. All organisms are afraid of their surrounding

IV. All organisms are increased by their mass

(1) I only

(2) II only

(3) II, III only

(4) All the above



8. Statement A : The sum total of all the chemical reactions occurring in our body is metabolism.

Statement B : Not all plants, animals, fungi and microbes exhibit metabolism.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



9. Statement A : Increase in mass and increase in number of individuals are twin characteristics of growth

Statement B : In majority of lower animals and plants, growth and reproduction are mutually exclusive events.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



10. Statement A : Yeast and hydra reproduce through sexual reproduction called budding

Statement B : In bacteria, unicellular algae or Amoeba, reproduction is synonymous with growth

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



11. Match the following

Column I

A. Diatoms

B. Euglena

C. Protozoan

D. Dinoflagellates

Column II

P. Red tides

Q. Polishing

R. Mixotrophic nutrition

S. Parasite

(1) A - R, B - P, C - Q, D - R

(2) A - P, B - R, C - S, D - Q

(3) A - Q, B - R, C - S, D - P

(4) A - S, B - R, C - Q, D - P



12. Match the following.

Column I

A. Phycomycetes

B. Basidiomycetes

C. Deuteromycetes

D. Ascomycetes

Column II

P. Fungi imperfecti

Q. Sac fungi

R. Club fungi

S. Algal fungi

(1) A - S, B - R, C - P, D - Q

(2) A - Q, B - S, C - R, D - P

(3) A - R, B - S, C - P, D - Q

(4) A - Q, B - P, C - R, D - S



13. Match the column I with Column II.**Column I****A. Viroids****B. Prions****C. Lichen****D. Virus****Column II****P. Symbiosis****Q. Obligate parasite****R. Mad cow disease****S. Potato- spindle Disease****(1) A - R, B - Q, C - P, D - S****(2) A - Q, B - R, C - P, D - S****(3) A - R, B - S, C - Q, D - P****(4) A - S, B - R, C - P, D - Q**

14. Which of these are viral diseases?

- (1) Polio, HIV
- (2) Influenza, small pox
- (3) Chicken pox, measles
- (4) All



15. Study the following characters carefully and give answer :-

- i. Bacterial structure is very simple because they are prokaryotic and unicellular**
- ii. Bacteria as a group show most extensive metabolic diversity**
- iii. Based on shape, bacteria are grouped into two categories only.**
- iv. Most of the bacteria do not have cell wall.**

- (1) i, ii-correct ; iii, iv-incorrect
- (2) iii, iv-correct ; i, ii-incorrect
- (3) i, iv-correct ; ii, iii-incorrect
- (4) ii, iii-correct ; i, iv-incorrect



16. Character belongs to slime moulds

A. Saprophytes

B. Forms plasmodium under unfavourable condition

C. Spores posses true walls

D. Spores are dispersed by water

E. Body moves along decaying twings and leaves engulfing organic matter.

(1) All except A, E

(2) All except B, D

(3) All except C and E

(4) All except B, E



17. Archaeobacteria and Cyanobacteria are characterized by common features i.e

- 1. Rigid cell wall**
- 2. Motile structure**
- 3. cellular thallus organisation**
- 4. Heterocyst for N_2 fixation**
- 5. Mucilagenous sheath**
- 6. Simple structure and complex in behaviour**

- (1) 1, 3 and 6
- (2) 1, 4 and 5
- (3) 1,2 and 3



(4) 1, 4 and 6

18. Who recognised certain microbes as causal organism of the mosaic disease?

- (1) Dmitri Ivanowsky
- (2) Dmitri Pasteur
- (3) Dmitri Beijerinck
- (4) Dmitri Stanelly



19. Bovine spongiform encephalopathy is caused by

- (1) Virus
- (2) Viroids
- (3) Fungi
- (4) Prions



20. Consider the following statements about Ascomycetes. Which one of the statement given below is false?

- (1) They are saprophytic, decomposer, coprophilous and parasitic
- (2) Include unicellular and multicellular forms
- (3) Mycelium is coenocytic and aseptate
- (4) Aspergillus, Claviceps, and Neurospora are important examples of Ascomycetes



21. In Basidiomycetes, the mycelium is

- (1) Branched and aseptate
- (2) Branched and septate
- (3) Unbranched and aseptate
- (4) Unbranched and septate



22. Assertion : Fungi consists of intervening stage between plasmogamy and Karyogamy called as dikaryotic stage and it contain two haploid nuclei.

Reason : Fungal cellulose or chitin is basically made up of acetyl glucosamine

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.



23. Prions posses

- (1) DNA only
- (2) Either DNA or RNA
- (3) Protein only
- (4) RNA only



24. Identify the agent which was similar in size to viruses?

- (1) Prions
- (2) Virus
- (3) Lichen
- (4) Mycorrhizae



25. Mad cow disease is caused by

- (1) Proteinaceous molecule
- (2) DNA
- (3) RNA
- (4) Both 2 and 3



26. Assertion : Prions are infectious proteins

Reason : Viroids are tiny, naked molecules of RNA

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.



27. Claviceps and Albugo are members of _____ respectively.

- (1) Ascomycetes, Phycomycetes
- (2) Basidiomycetes, Phycomycetes
- (3) Zygomycetes, Phycomycetes
- (4) Phycomycetes, Basidiomycetes



28. The following features belong to class

A. Members are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants

B. Fusion of gametes may be isogamous, anisogamous or oogamous.

C. Mycelium is aseptate and coenocytic.

D. Spores are endogenously produced in sporangium.

E. Asexual reproduction by zoospores or aplanospores.

(1) Ascomycetes

(2) Deuteromycetes

(3) Phycomycetes

(4) Basidiomycetes



29. Which one of the following is not a criterion for classification of fungi?

- (1) Morphology of mycelium
- (2) Mode of nutrition
- (3) Mode of spore formation
- (4) Fruiting body



30. Choose the correct statements related to Fungi.

I. They show great diversity and morphology

II. They cause orange rot.

III. Toadstools are edible fungi.

IV. White spots on mustard leaves is due to parasitic fungus.

(1) I, II, III, IV

(2) I, III, IV

(3) I, II, IV

(4) II, III, IV



31. Sexual reproduction is of isogamous, anisogamous and oogamous in

- (1) Phaeophyceae and Rhodophyceae
- (2) Rhodophyceae
- (3) Chlorophyceae and Phaeophyceae
- (4) Chlorophyceae and Rhodophyceae

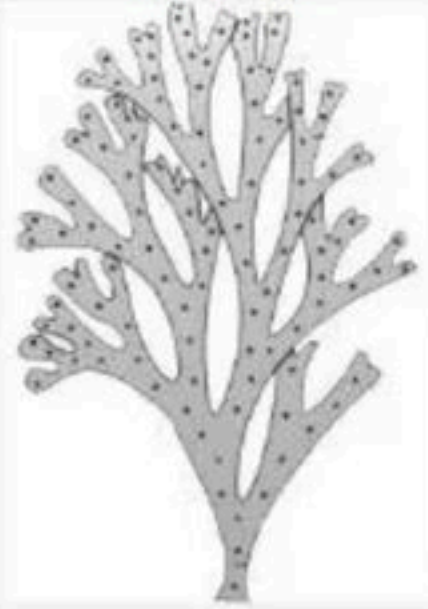


32. Which of the following statements is true?

- (1) Trimerous condition of floral whorls is characteristic of dicotyledons
- (2) Adiantum is also called flying fern
- (3) In gymnosperms the vascular system consists of xylem without vessels and phloem with companion cells
- (4) Riccia and Marchantia are liverworts.



33. Name the diagram given below_



(1) Kelps

(2) Polysiphonia

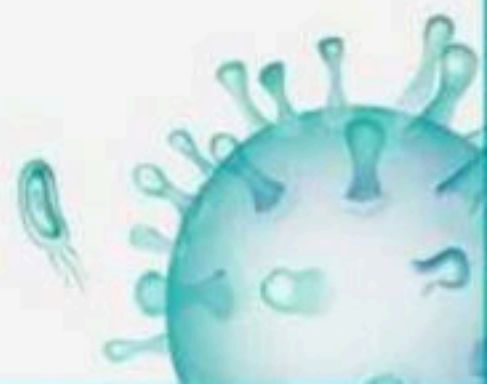
(3) Laminaria

(4) Dictyota



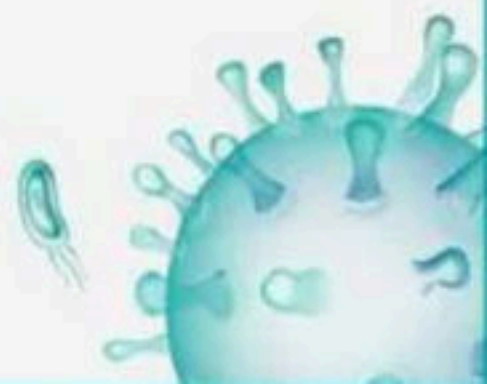
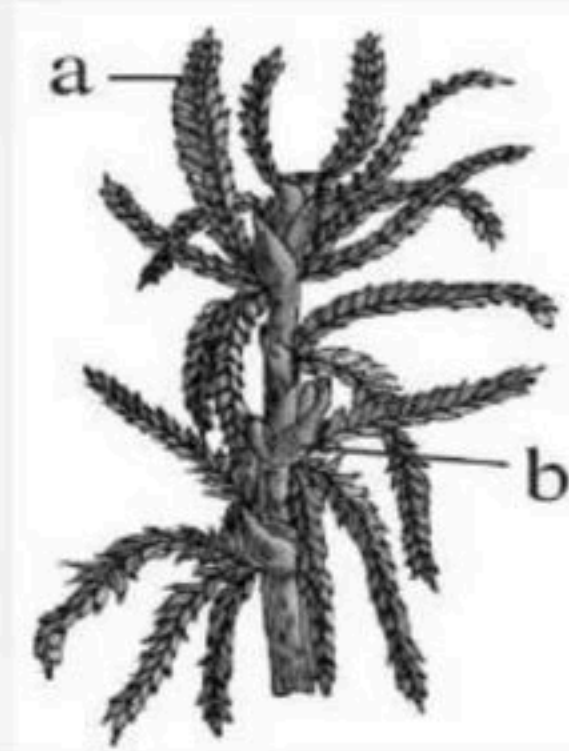
34. Which of the following statement is incorrect regarding bryophytes?

- (1) Water is essential for sexual reproduction
- (2) The main plant body is haploid
- (3) They usually occur in damp, humid and shady localities.
- (4) The presence of autotrophic, independent sporophyte



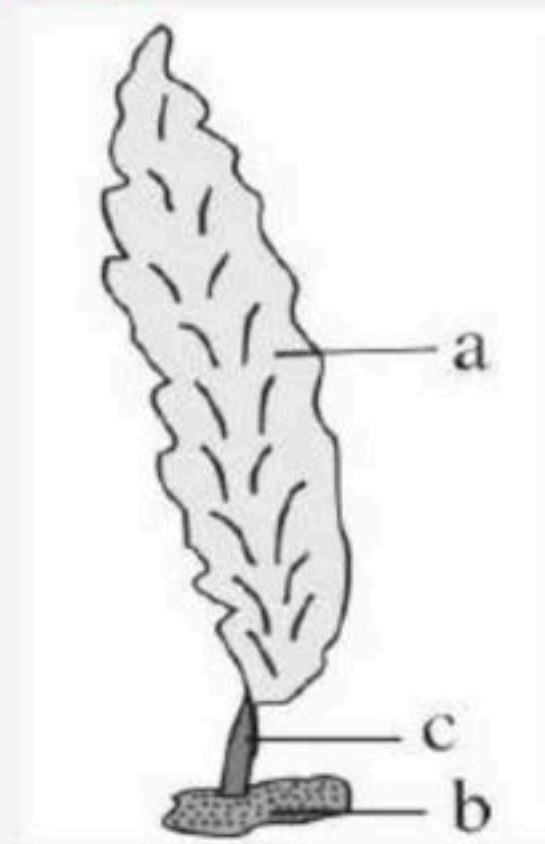
35. Identify the following plant given in the diagram.

- (1) Funaria gametophyte
- (2) Sphagnum gametophyte
- (3) Marchantia male gametophyte
- (4) Marchantia female gametophyte.



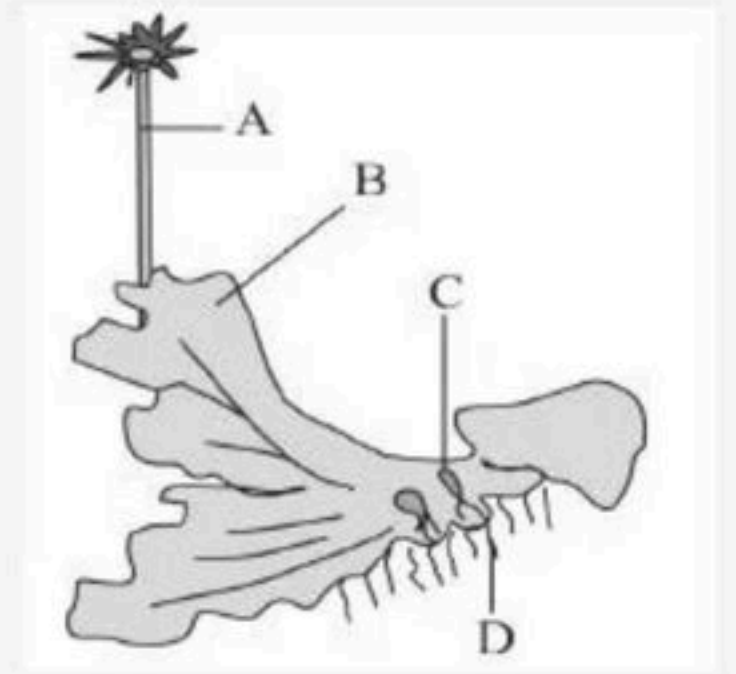
36. Label the diagram of Laminaria a, b and c respectively are_

- (1) a-Stipe, b-frond, c-holdfast
- (2) a-Holdfast, b-rhizoid, c-lamina
- (3) a-Frond, b-holdfast, c-stipe
- (4) a-Frond, b-stripe, c-holdfast



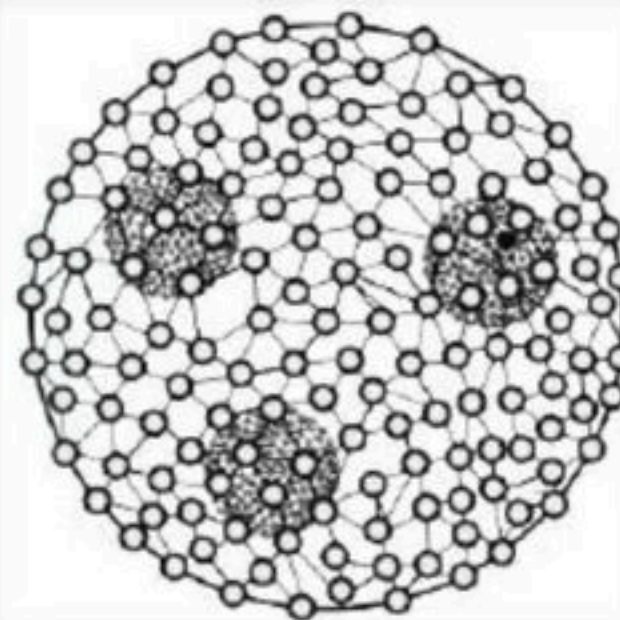
37. Examine the figure given below and select the correct option giving all the four parts (A, B, C, D) rightly identified:

- (1) A - Archegoniophore, B - Female thallus, C – Gemma cup, D - Rhizoids
- (2) A - Archegoniophore, B - Female thallus, C - Bud, D - Food
- (3) A - Seta, B - Sporophyte, C - Protonema, D - Rhizoids
- (4) A - Archegoniophore, B - Male thallus, C - Globule, D - Roots



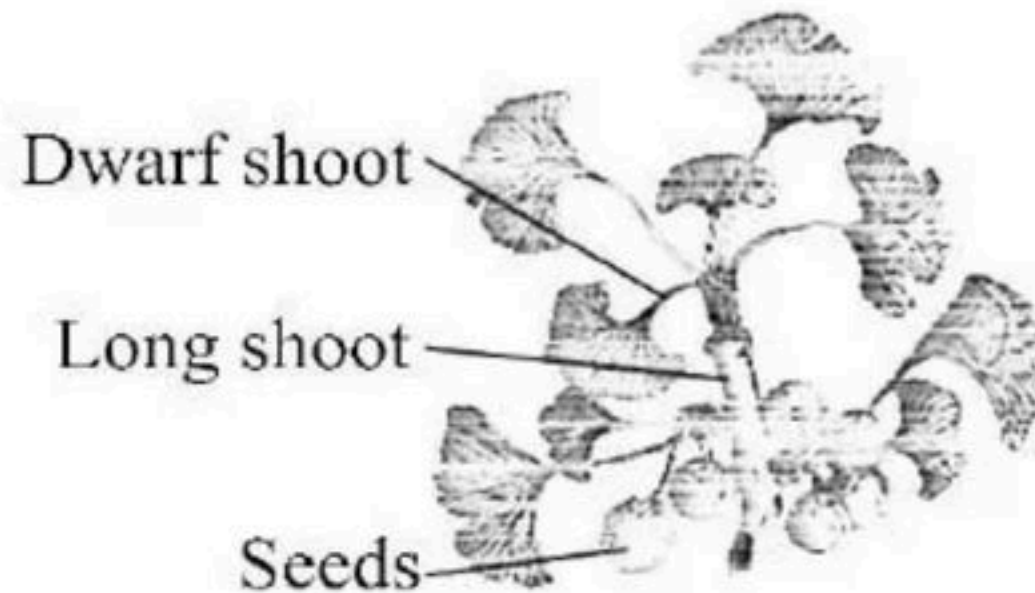
38. Reproduction in species given below is_

- (1) Oogamous
- (2) Anisogamous
- (3) Isogamous
- (4) None of the above



39. Roots in the following species are_

- (1) Fibrous roots
- (2) Stilt roots
- (3) Tap roots
- (4) Coralloid roots



40. Statement A : In red algae sexual reproduction is accompanied by complex pre-fertilisation development.

Statement B : Porphyra, Polysiphonia, Gracilaria are the members of red algae.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



41. Statement A : Most of the red algae have complex body organization.

Statement B : In red algae food is stored as amylopectin and glycogen.

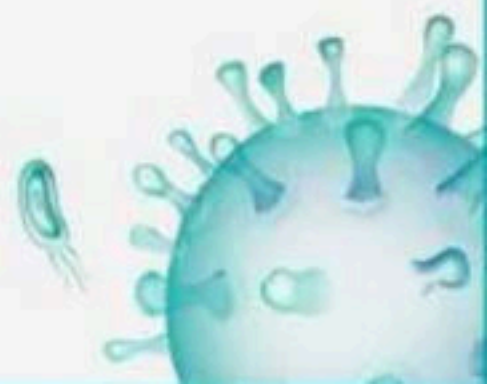
- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



42. Statement A : Prothallus is conspicuous in pteridophyta.

Statement B : Spread of living pteridophytes is limited.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



43. Which of the following statements with respect to algae are correct.

- I. Fusion between one large, non-motile female gamete and a smaller, motile male gamete is termed as oogamous.**
- II. Fusion of two gametes similar in size is termed as oogamous.**
- III. Fusion of two gametes similar in size is called anisogamous.**
- IV. In Chlorophyceae, the major pigments are chlorophyll a and b, and the food is stored as starch.**
- V. In rhodophyceae, the major pigments are chlorophyll a and d, and the food is stored as mannitol.**

(1) I and V

(2) III and V

(3) I and II

(4) I and IV



44. Choose the incorrect statements for the sporophyte of bryophytes.

I. Sporophyte is multicellular, free-living, and provides nourishment to the gametophyte.

II. Some cells of the sporophyte undergo meiosis to produce haploid spores.

III. These spores germinate to produce sporophyte.

(1) I and II

(2) I and III

(3) II and III

(4) I, II and III



45. Statement A : Gemmae are green, multicellular and sexual buds

Statement B : Gemma develop small receptacle called as gemma cups

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



46. Match the columns I and II.

Column I

A. Gorgonia

B. Meandrina

C. Pennatula

D. Adamsia

Column II

P. Brain coral

Q. Sea-fan

R. Sea-pen

S. Sea anemone

(1) A – R, B – P, C – Q, D – S (2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S (4) A – Q, B – P, C – R, D – S



47. Match the columns I and II.

Column I

A. Cucumaria

B. Arthropoda

C. Annelida

D. Mollusca

Column II

P. Book gills

Q. Moist cuticle with skin

R. Water vascular system

S. Gills

(1) A – R, B – P, C – Q, D – S (2) A – S, B – R, C – P, D – Q

(3) A – P, B – R, C – Q, D – S (4) A – Q, B – P, C – R, D – S



48. Which of the following animal groups is not correctly paired with its characteristic?

- (1) Arthropoda-Chitinous exoskeleton
- (2) Mollusca - Exoskeleton is absent
- (3) Echinodermata-Excretory system is absent
- (4) Hemichordata- Respiration through gills



49. Match the columns I and II.

Column I

A. Chelone

B. Testudo

C. Chameleon

D. Calotes

E. Hemidactylus

Column II

P. Tree lizard

Q. Turtle

R. Garden lizard

S. Wall lizard

T. Tortoise

(1) A – R, B – P, C – Q, D – T, E – S (2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q (4) A – Q, B – T, C – P, D – R, E – S



50. Match Column I with Column II :

Choose the correct answer from the options given below :

Column I

A. Pterophyllum

B. Myxine

C. Pristis

D. Exocoetus

Column II

P. Hag fish

Q. Saw fish

R. Angel fish

S. Flying fish

(1) A – Q, B – P, C – R, D – S (2) A – R, B – P, C – Q, D – S

(3) A – S, B – P, C – Q, D – R (4) A – R, B – Q, C – P, D – S



51. Statement A : A bud is present in the axil of petiole in both simple and compound leaves, and in the axil of leaflets of the compound leaf.

Statement B : In a palmately compound leaf a number of leaflets are present on a common axis, the rachis,

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



52. How many statements are correct for opuntia and euphorbia

(i) Stem is modified into leaf like organs

(ii) Leaf are modified into spine

(1) i is correct

(2) ii is correct

(3) Both are correct

(4) None are correct



53. Select the correct answer

- (1) Placentation -Arrangement of ovules
- (2) Phyllotaxy- arrangement of leaves at nodes
- (3) Venation - arrangement of veins
- (4) All are correct



54. Match the columns I and II.

Column I

A. Monoadelphous

B. Diadelphous

C. Polyadelphous

(1) A – R, B – P, C – Q

(3) A – P, B – R, C – Q

Column II

P. Citrus

Q. China rose

R. Pea

(2) A – Q, B – R, C – P

(4) A – Q, B – P, C – R



55. Read the following statements and find out the incorrect statement.

- (1) In potato, Colocasia and ginger the underground stem act as organs of perennation to tide over conditions unfavourable for growth
- (2) Underground stem of potato, ginger and turmeric are modified to store food in them
- (3) Stem tendrils help cucumber and Bougainvillea plants to climb
- (4) The region of stem where leaves are born are called nodes while internodes are portions between two nodes



56. Which of the following is incorrect?

- (1) Roots help in water and mineral absorption from soil
- (2) Roots provide a proper anchorage
- (3) Roots synthesise plant growth regulators
- (4) Roots are always positively geotropic



57. Identify the different type of placentation?

i. Axile, Central

ii. Free central, Basal

iii. Marginal, parietal

(1) Only i

(2) Only i, ii

(3) Only i, iii

(4) All are correct



58. In which type of placentation, the ovules develop on the inner wall of the ovary or on peripheral part?

- (1) Marginal placentation
- (2) Parietal placentation
- (3) Axile placentation
- (4) Free-central placentation



59. Which of the following statement(s) is/are correct about the fruit?

I. Fruit is a mature or ripened ovary, developed before fertilization.

II. It consists of a wall or pericarp and seeds.

III. When pericarp is thick and fleshy, it is differentiated into outer mesocarp, middle epicarp and inner endocarp.

IV. In mango and coconut, the fruit is known as a berry.

(1) Only I

(2) Both II and III

(3) Only II

(4) All of these



60. Select the type of In fabaceae, the corolla is papilionaceous, consisting of a _____ standard, two _____ wings, two _____ ones forming a keel

- (1) Posterior, medial, anterior
- (2) Anterior, medial, anterior
- (3) Posterior, lateral, anterior
- (4) Anterior, lateral, posterior



61. Match the columns I and II.

Column I

A. Variation in length of filaments within flower

B. Apocarpous

C. Syncarpous

Column II

P. Lotus

Q. Salvia

R. Tomato

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



62. Identify the incorrect match

- (1) Zygomorphic - bean
- (2) Asymmetric - Canna
- (3) Asymmetric - Cassia
- (4) Actinomorphic - Datura



63. Find the correct floral formula of actinomorphic, bisexual flower having five united sepals, five united petals, epipetalous five stamens and bicarpellary syncarpous superior ovary

$$(1) \oplus \hat{\bigcirc}_{\perp} K_{(5)} \overbrace{C_{(5)} A_5} G_{(2)}$$

$$(2) \oplus \hat{\bigcirc}_{\perp} K_{(5)} C_{(5)} A_5 G_{(2)}$$

$$(3) \oplus \hat{\bigcirc}_{\perp} K_{(3)} C_{(3)} A_{(5)} G_{(2)}$$

$$(4) \oplus \hat{\bigcirc}_{\perp} K_{(5)} \overbrace{C_{(5)} A_5} G_{(2)}$$



64. Statement A : The plumule and radicle are enclosed in sheaths which are called coleorhiza and coleoptile respectively.

Statement B : In monocotyledonous seed, embryo consists of scutellum and short axis with plumule and radicle.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



65. Select the incorrect statement from the following:

- (1) In monocot leaf, the base expands into a sheath covering the stem partially or wholly.
- (2) In all leguminous plants, the leaf base may become swollen, which is called the pulvinus.
- (3) The lamina or leaf blade is the green expanded part of the leaf with veins and veinlets.
- (4) All are correct



66. Tap roots of which of the following get swollen and store food?

- (1) *Daucus carota*
- (2) *Brassica rapa*
- (3) Both (1) and (2)
- (4) *Ipomoea batatas*



67. Read the following statements carefully and select the correct statement____

- (1) In plants such as bean, gram and pea the endosperm is not present.
- (2) The plumule and radicle surrounds the sheath called coleoptile and coleorhiza respectively.
- (3) The outer covering of scutellum separates embryo from endosperm is called as aleurone layer.
- (4) The cotyledons of dicots are called as scutellum individually.



68. Leaves become modified into spines in:

- (1) Pea
- (2) Onion
- (3) Silk Cotton
- (4) Opuntia



69. Statement A : During flower formation, the apex produces different kinds of floral appendages laterally at successive nodes along with leaves

Statement B : When a shoot tip transforms into a flower, it is sometimes solitary.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



70. Leaves of dicotyledonous plants generally possess_____ venation, while parallel venation is the characteristic of_____ monocotyledons

- (1) Reticulate, all
- (2) Parallel, most
- (3) Reticulate, most
- (4) Parallel, all



71. The cells which becomes structurally and functionally specialized and lose the ability to divide are known as

- (1) Permanent cells
- (2) Mature cells
- (3) Both (1) and (2)
- (4) Complex tissue



72. Statement A : Cuticle is absent in roots.

Statement B : Stomata are structures present in the endodermis of leaves.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



73. Which one of the following represents the characters of epidermal cells?

- (1) They are parenchyma cells
- (2) Presence of small amount of cytoplasm
- (3) Large vacuole
- (4) All of these



74. Match the following and choose the correct option from below

Column I

A. Cuticle

B. Bulliform cells

C. Stomata

D. Epidermis

(1) A-R, B-S, C-P, D-Q

(3) A-R, B-Q, C-S, D-P

Column II

P. Guard cells

Q. Single layer

R. Waxy layer

S. Empty colourless cell

(2) A-P, B-Q, C-R, D-S

(4) A-R, B-Q, C-P, D-S



75. Medullary rays are present in

- (1) Dicot root
- (2) Monocot root
- (3) Monocot stem
- (4) Dicot stem



76. Statement A : The vascular bundles are surrounded by a layer of thick walled bundle sheath cells, in dicotyledonous leaf.

Statement B : The mesophyll is not differentiated into palisade and spongy parenchyma in isobilateral leaf.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.

77. On what basis we classify sclerenchyma into fibres or



sclereids?

- (1) Structure
- (2) Origin
- (3) Development
- (4) All of these

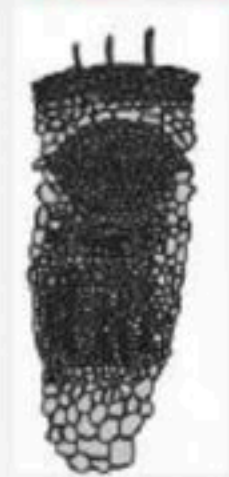


78. The tissue system in plants is divisible into three types on the basis of

- (1) Their structure
- (2) Their size
- (3) Their location in plant body
- (4) Both 1 and 3



79. Carefully observe the given diagram and select the correct option.



- (1) It is section of monocot root.
- (2) It is section of dicot stem
- (3) It is section of monocot stem
- (4) It is section of dicot root



80. Statement A : The collenchyma performs various functions like photosynthesis, storage, secretion.

Statement B : The collenchyma occurs in layers below the epidermis in most of the dicotyledonous plants.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



81. Statement A : In dicotyledonous stems, cambium surrounds phloem and xylem.

Statement B : Cambium possess the ability to form secondary xylem and phloem tissues.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



82. Statement A : Phloem transports food materials, usually from leaves to other parts of the plant.

Statement B : Gymnosperms lack vessels, sieve tubes and companion cells.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



83. Statement A : In roots, the protoxylem lies towards the centre (pith) and the metaxylem lies towards the periphery of the organ.

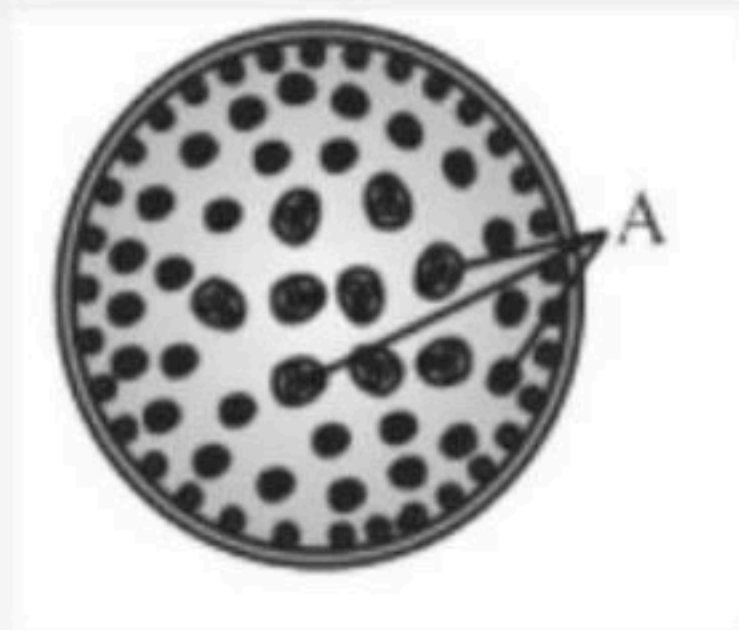
Statement B : In stems, the protoxylem lies towards periphery and metaxylem lies towards the centre.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



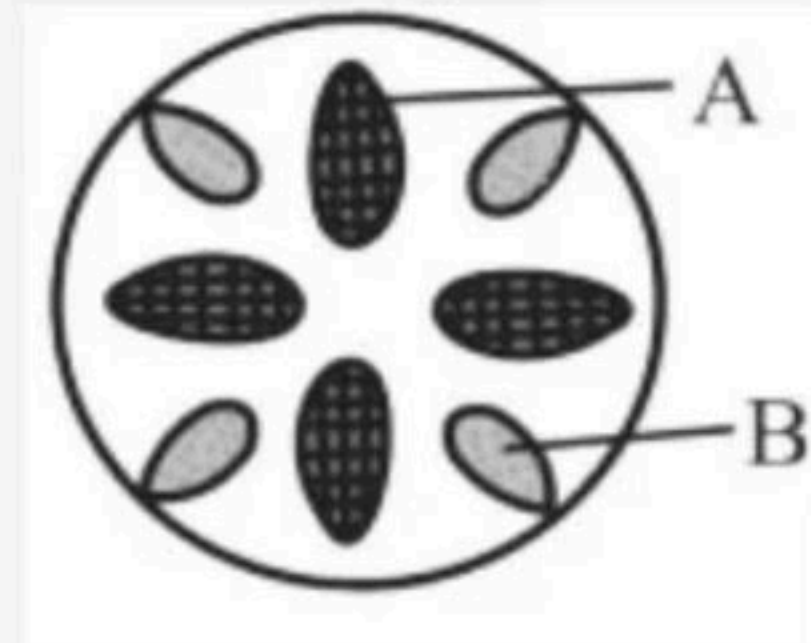
84. The given diagram is transverse section of monocot stem. Identify labeled A in the given diagram?

- (1) Pith
- (2) Medullary rays
- (3) Vascular bundles
- (4) Starch sheath



85. The given diagram is of vascular bundle. Identify the component which transports water and food respectively in the plant?

- (1) Only A
- (2) Only B
- (3) A, B
- (4) B, A



86. Tendons attach

- (1) Bones to bones
- (2) Muscle to bone
- (3) Muscles to muscles
- (4) Cartilage to bone



87. Read the following statements about cockroach

I. In male cockroach, a characteristic mushroom shaped gland is present in the 6th – 7th abdominal segments which functions as an accessory reproductive gland.

II. Cockroach is uricotelic.

III. The fat body and uricose gland help in excretion.

IV. Blood from sinuses enter into heart through ostia and is pumped anteriorly to sinuses again.

Which of the above statements are correct?

(1) II and III

(2) I, II and IV

(3) I, II, III and IV

(4) I and IV



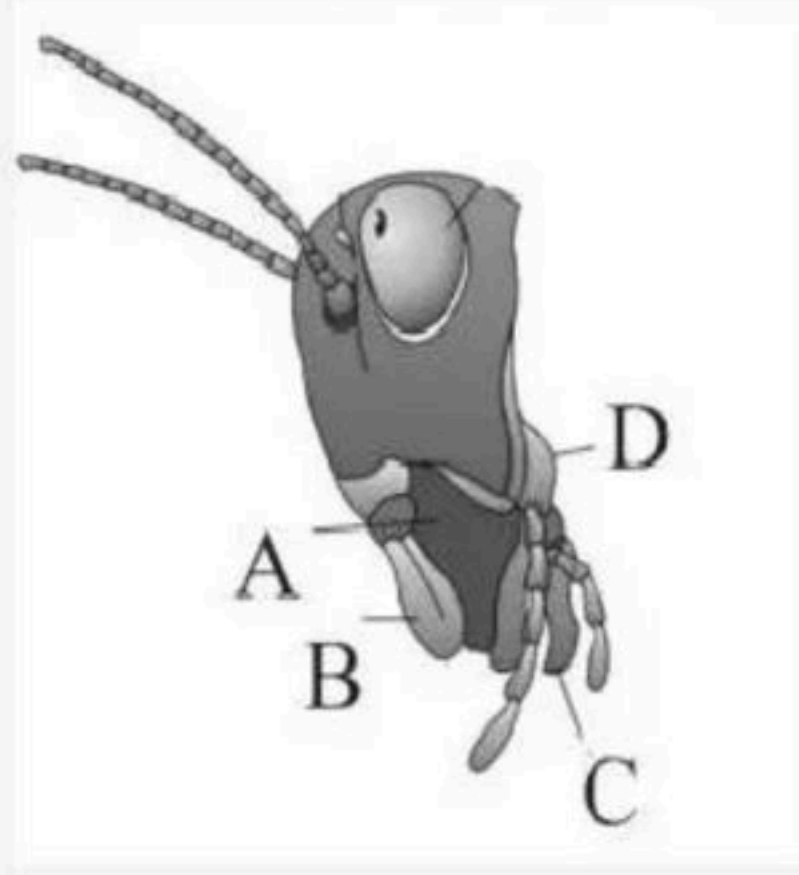
88. Identify maxilla and mandible in the diagram

(1) A, B

(2) D, B

(3) D, A

(4) A, D



89. Statement A : The RBCs, WBCs and platelets are enucleated in frogs.

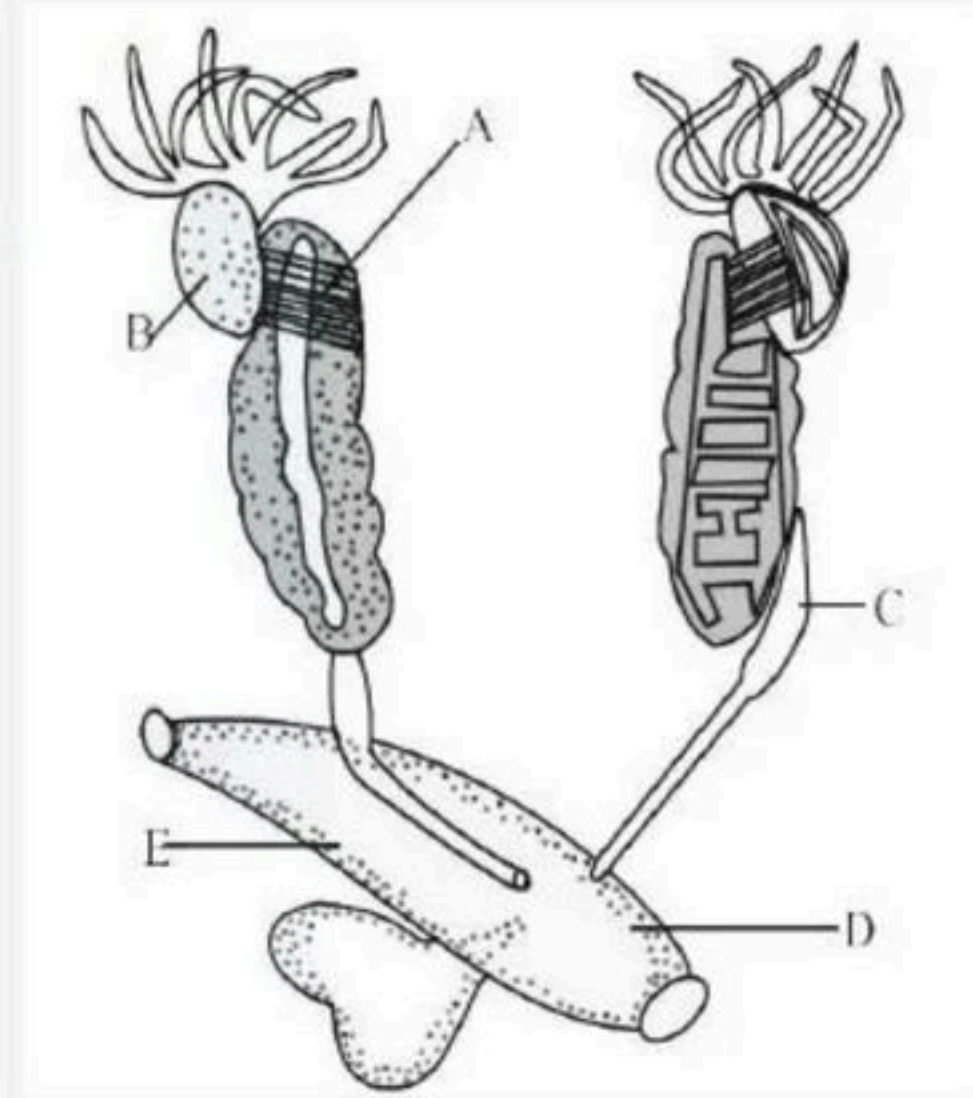
Statement B : Frogs, are omnivores and nocturnal

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



90. Given below is the diagrammatic sketch of male reproductive system of frog. Select the option that correctly identifies one of the parts along with its function.

- (1) A: Bidder's canal - transports sperm
- (2) : Adrenal gland - exocrine gland
- (3) : Urinogenital duct - transports both urine and sperm
- (4) D : Urinary bladder - stores urine



91. The columnar epithelium in human body is found in

- (1) Stomach
- (2) Lungs
- (3) Kidney
- (4) Fallopian tube



92. Which of the following is known as tegmina in cockroach?

- (1) Fore wing
- (2) Hind wing
- (3) Metathorax
- (4) Prothorax



93. Select the correct classification of *Rana tigrina* upto genus.

- (1) Chordata - Gnathostomata - Amphibia - Rana
- (2) Chordata - Amphibia Gnathostomata - tigrina
- (3) Chordata - Ambhibia Gnathostomata - Rana
- (4) Amphibia - Chordata - Gnathostomata - Rana



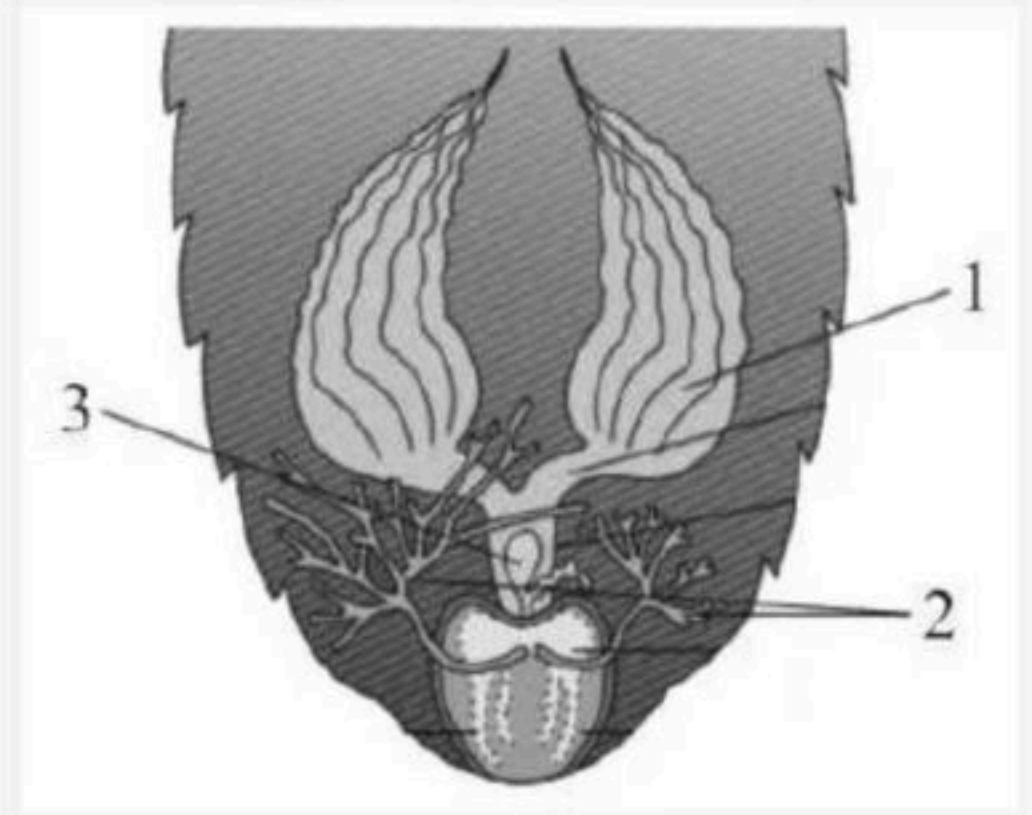
94. In which of the following structure in the cockroach six highly chitinous teeth are present?

- (1) Crop
- (2) Proventriculus
- (3) Intestine
- (4) Oesophagus



95. Function of (1) in the given diagram representing reproductive system of cockroach is

- (1) 1-Ovary 2-Phallic glands 3-Spermatheca
- (2) 1- Ovary 2- Collateral glands 3- Spermatophores
- (3) 1- Oviduct 2- Collateral glands 3- Spermatheca
- (4) 1- Ovary 2- Collateral glands 3- Spermatheca



96. In frogs, sensory papillae, taste buds and nasal epithelium are ____ while eyes and internal ears are well-organised structures.

- (1) Cellular aggregations around epithelium
- (2) Cellular aggregation around bone
- (3) Cellular aggregations around muscle
- (4) Cellular aggregations around nerve endings



97. Match Column I with Column II related to digestive system of cockroach.
Choose the correct answer from the options given below:

Column I		Column II	
A	The structures used for storing of food	P	Gizzard
B	Ring of 6-8 blind tubules at junction of foregut and midgut	Q	Gastric Caecae
C	Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut	R	Malpighian tubules
D	The structures used for grinding the food	S	Crop

(4) A – S, B – Q, C – R, D – P

(4) A – P, B – Q, C – R, D – S

(4) A – S, B – R, C – Q, D – P

(4) A – R, B – Q, C – S, D – P



98. Match the columns I and II

Column I

A. Unicellular gland

B. Multicellular exocrine gland

C. Endocrine gland

Column II

P. Thymus

Q. Goblet cells

R. Salivary gland

(1) A – R, B – P, C – Q

(2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



99. Match Column- I with Column- II.

Choose the correct answer from the options give below:

Column I		Column II	
A	Mast cells	P	Ciliated epithelium
B	Inner surface of bronchiole	Q	Areolar connective tissue
C	Blood	R	Cuboidal epithelium
D	Tubular parts of nephron	S	Specialised connective tissue

(1) A-P, B-Q, C-S, D-R

(2) A-Q, B-R, C-P, D-S

(3) A-Q, B-P, C-S, D-R

(4) A-R, B-S, C-Q, D-P



100. Match the columns I and II in the frog.

Column I

A. Forebrain

B. Midbrain

C. Hind-brain

Column II

P. A pair of optic lobes

Q. A pair of olfactory lobes

R. Cerebellum

(1) A – R, B – P, C – Q

(3) A – P, B – R, C – Q

(2) A – Q, B – R, C – P

(4) A – Q, B – P, C – R



101. Statement A : In plant cells the vacuoles can occupy up to 10 per cent of the volume of the cell.

Statement B : In Amoeba the contractile vacuole is important for osmoregulation and excretion.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



102. Kinetochore is

- (1) Centromere
- (2) Protein discs at primary constriction
- (3) Protein discs at secondary constriction
- (4) DNA at centromere



103. Which of the following is the correct statement about cellwall of eukaryotic cell?

- (1) Algae have cell wall, made of cellulose, galactans, mannans and minerals like calcium carbonate,
- (2) It mainly consists of Ca-pectate
- (3) The cell wall and middle lamellae may be traversed by plasmodesmata
- (4) All of the above



104. Statement A : Cell membrane is mainly composed of lipids and proteins.

Statement B : The major lipids are phospholipids that are arranged in a monolayer.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



105. Statement A : Ribosomes are membrane bound organelles.

Statement B : Ribosomes are found in all cells- both eukaryotic as well as prokaryotic.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



106. Statement A : Thylakoids are arranged in stacks like the piles of coins called grana.

Statement B : The membrane of the thylakoids enclose a space called a lumen

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



107. Statement A : In prokaryotes, ribosomes are associated with the plasma membrane of the cell.

Statement B : Several mRNAs may attach to a single ribosome and form a chain called polyribosomes or polysome

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



108. Match the columns I and II and choose the correct option

Column I

A Amyloplasts

B Elaioplasts

C Aleuroplasts

D Flattened membranous sacs

Column II

P Oils and fats

Q Thylakoids

R Carbohydrates

S Proteins

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-P, C-Q, D-S

(3) A-R, B-P, C-S, D-Q

(4) A-S, B-Q, C-P, D-R



109. Match the columns I and II and choose the correct option.

Column I

A Peripheral proteins

B 52 % of plasma membrane

C Integral proteins

D 40 % of plasma membrane

Column II

P Lipids

Q On the surface

R Protein

S Partially or totally buried

(1) A-Q, B-R, C-S, D-P

(3) A-R, B-Q, C-S, D-P

(2) A-Q, B-R, C-P, D-S

(4) A-R, B-P, C-Q D-S



110. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Chromosome with centromere slightly away from the middle	P	Acrocentric
B	Chromosome with a terminal centromere	Q	Metacentric
C	Chromosome with centromere is situated close to its end	R	Telocentric
D	Chromosome with middle centromere	S	Sub metacentric

(1) A-S, B-R, C-P, D-Q

(2) A-R, B-S, C-Q, D-P

(3) A-P, B-Q, C-S, D-R

(4) A-Q, B-P, C-R, D-S



111. Assertion : Cyanobacteria are photosynthetic, blue green algae with all the prokaryotic structures.

Reason : They are green due to presence of chloroplasts.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false



112. Assertion : Schleiden and Schwann together formulated the Cell theory.

Reason : Cells are formed from pre-existing cells.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false



113. Assertion : Mitochondria are common to both plant and animal cells.

Reason : The number of mitochondria in a cell doesnot corresponds to the function of the cell.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false



114. Assertion : The fimbriae are elongated tubular structures made of special protein.

Reason : The pili are small bristle like fibres sprouting out of the cell.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false



115. Following diagram shows different types of cells (1-6).

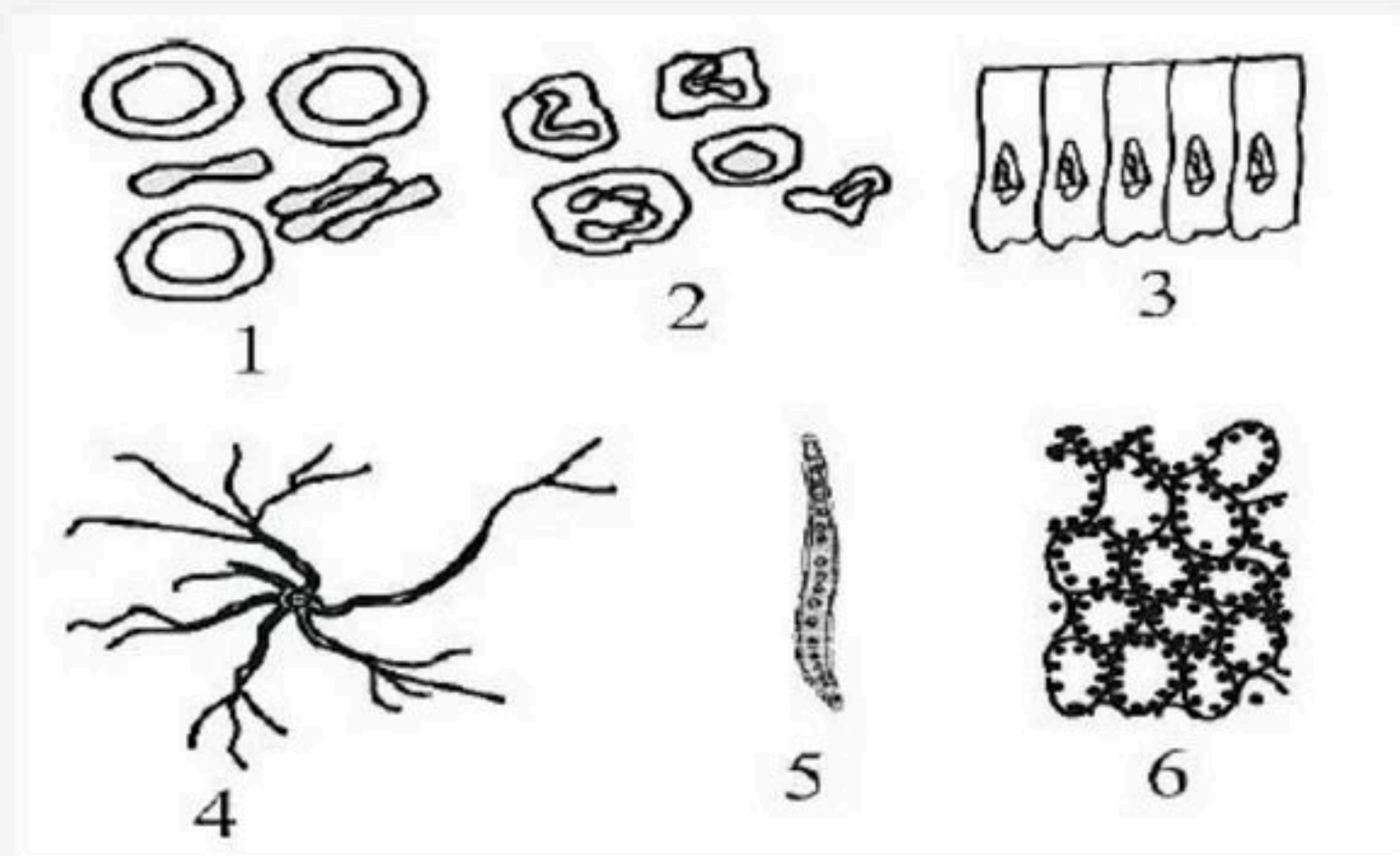
How many cells are present in animals and plants respectively?

(1) 3, 3

(2) 2, 4

(3) 1, 5

(4) 4, 2



116. Identify the different types of organism (A-D) in the given diagram?

(1) A - Typical eukaryotic cell

C – PPLO

(2) A - Typical eukaryotic cell

C – Viruses

(3) A – PPLO

C - Typical eukaryotic cell

(4) A – Viruses

C – PPLO

B - Typical bacteria

D - Viruses

B - Typical bacteria

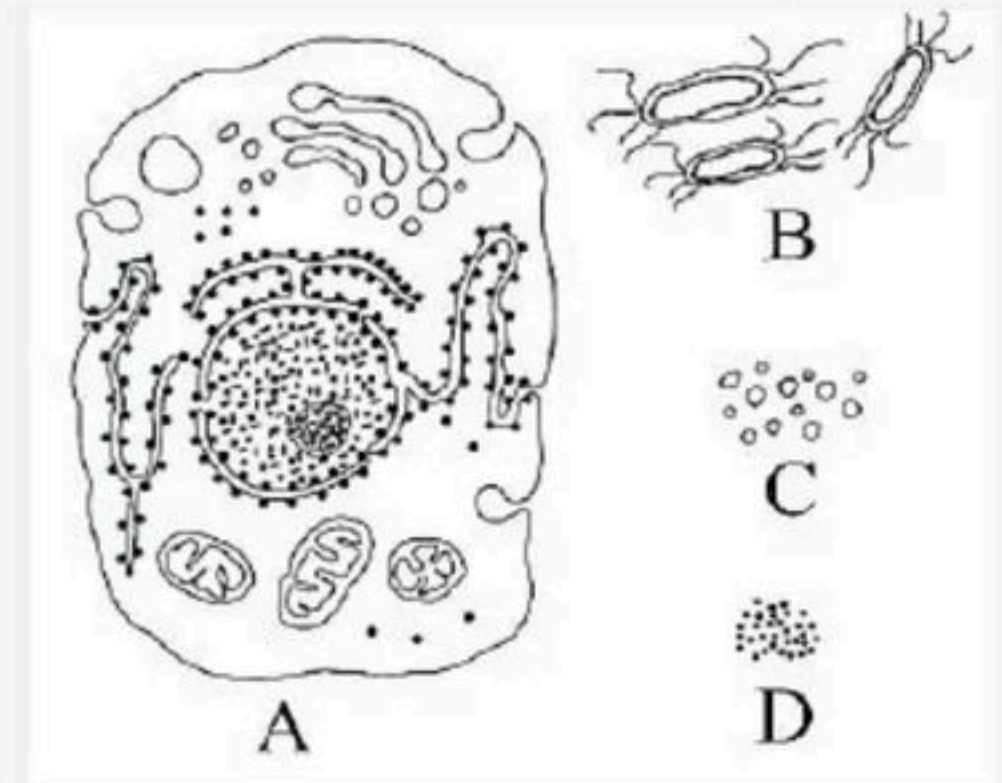
D - PPLO

B - Viruses

D - Typical bacteria

B - Typical bacteria

D - Typical eukaryotic cell



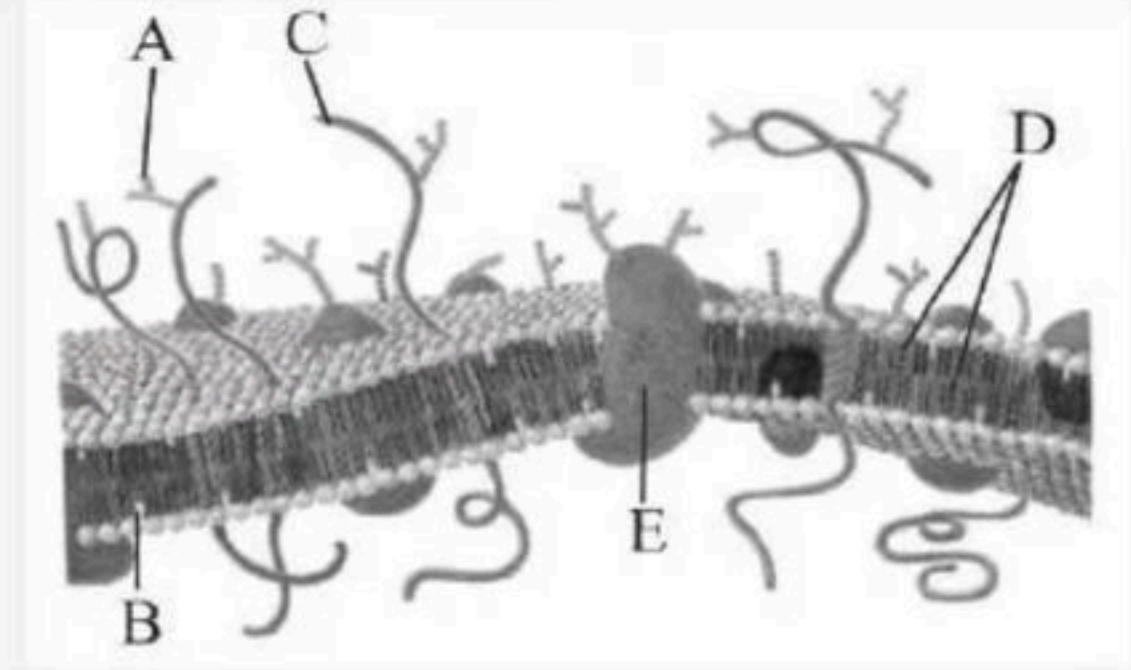
117. Identify the components labeled A – E in the given diagram?

(1) A - Sugar, B - Cholesterol, C - Peripheral Protein, D - Phospholipid bilayer, E - Integral protein

(2) A - Sugar, B - Cholesterol, C - Phospholipid bilayer, D - Peripheral Protein, E - Integral protein

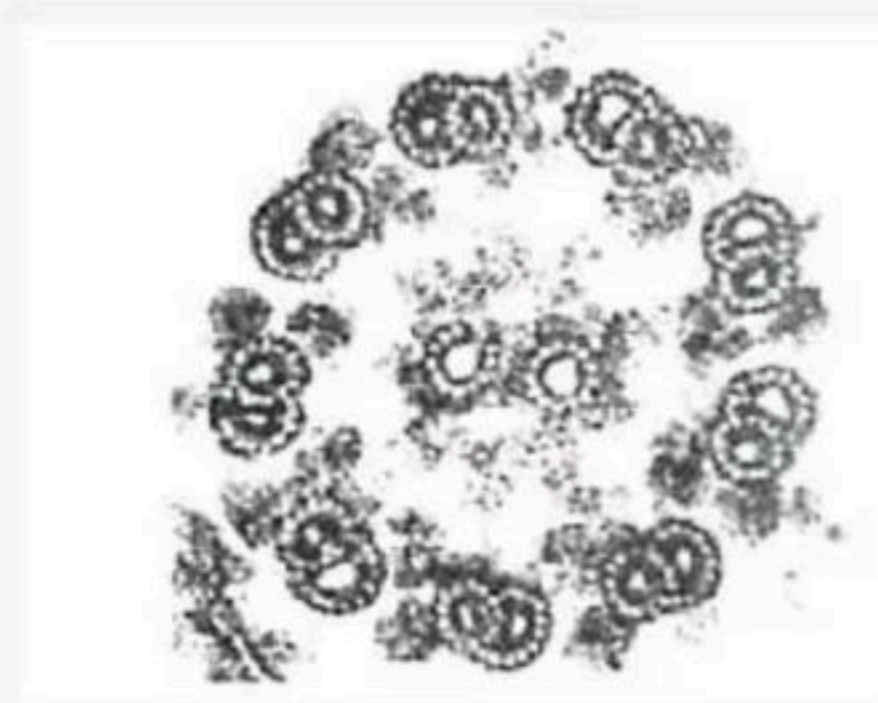
(3) A - Sugar, B - Phospholipid bilayer, C - Cholesterol, D - Peripheral Protein, E - Integral protein

(4) A - Integral protein, B - Sugar, C - Cholesterol, D - Peripheral Protein, E - Phospholipid bilayer



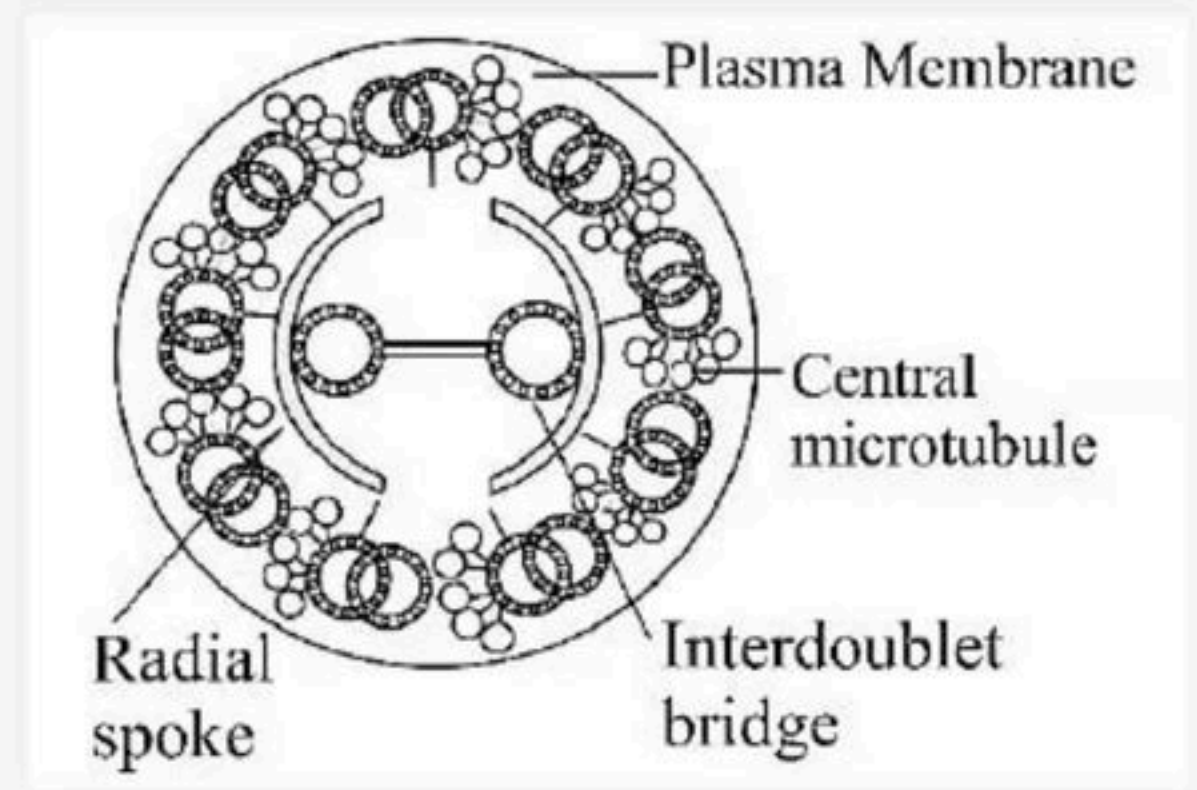
118. Carefully observe the given diagram and select the correct option.

- (1) It is section of mitochondria
- (2) It is section of chloroplast.
- (3) It is section of nucleus.
- (4) It is section of cilia or flagella.



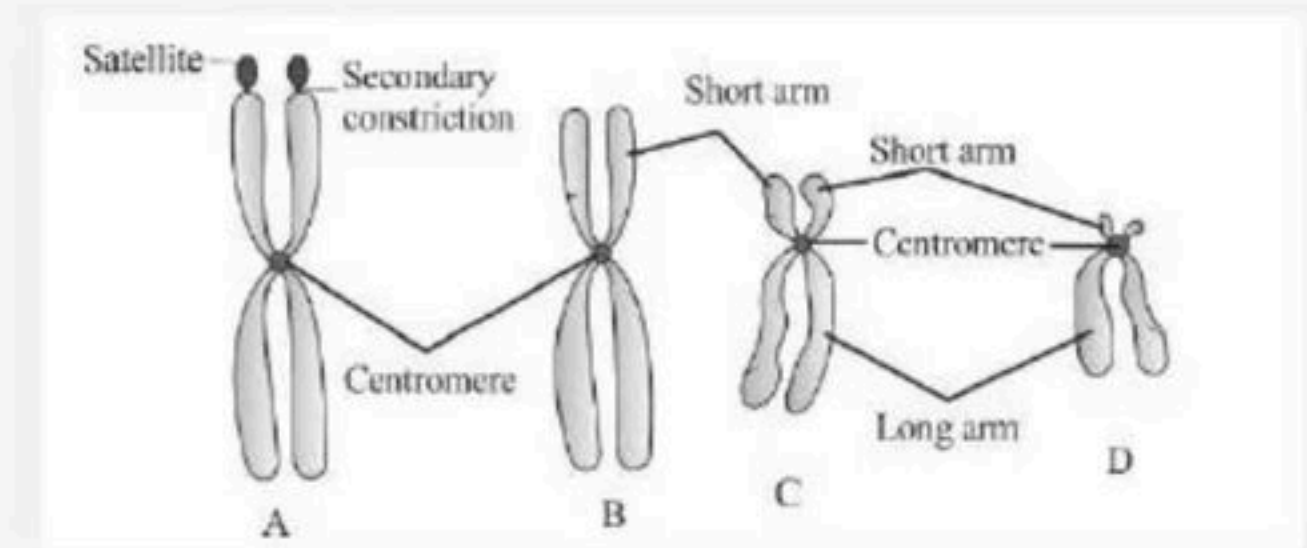
119. In the below diagram, identify the part which connects the peripheral microtubules to the central sheath.

- (1) Plasma membrane
- (2) Interdoublet bridge
- (3) Central microtubule
- (4) Radial spoke



120. Identify the components labeled A–D in the given chromosomes diagram?

- (1) A - sub-metacentric, B - metacentric, C - acrocentric, D - telocentric
- (2) A- metacentric, B - acrocentric, C - sub-metacentric, D - telocentric
- (3) A - metacentric, B-sub-metacentric, C - acrocentric, D - telocentric
- (4) A- acrocentric, B-sub-metacentric, C - metacentric, D - telocentric



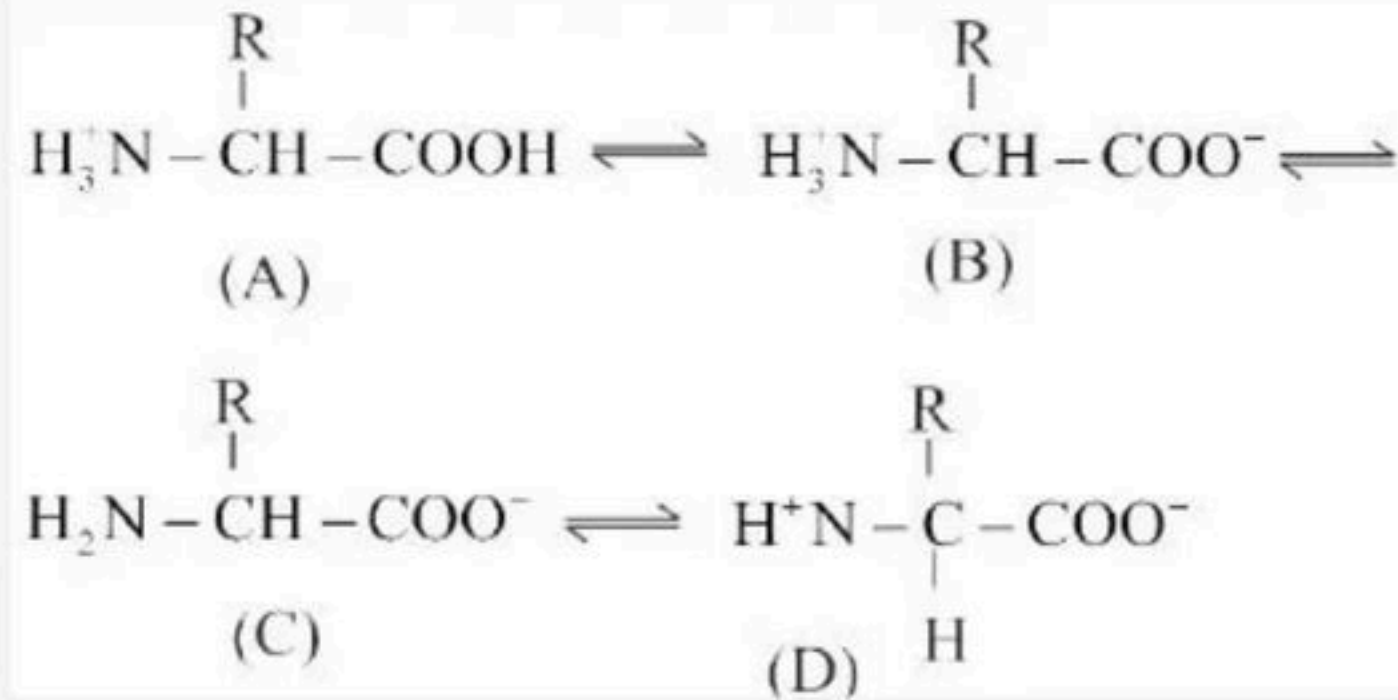
121. Identify the Zwitterionic form in the following diagram?

(1) A

(2) B

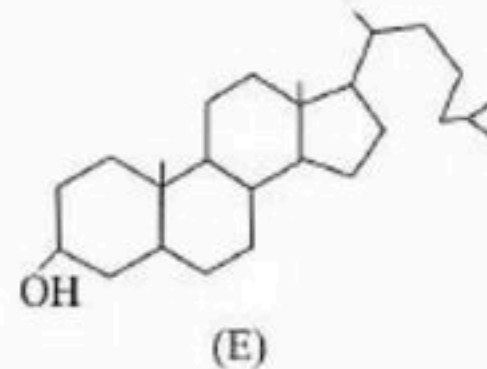
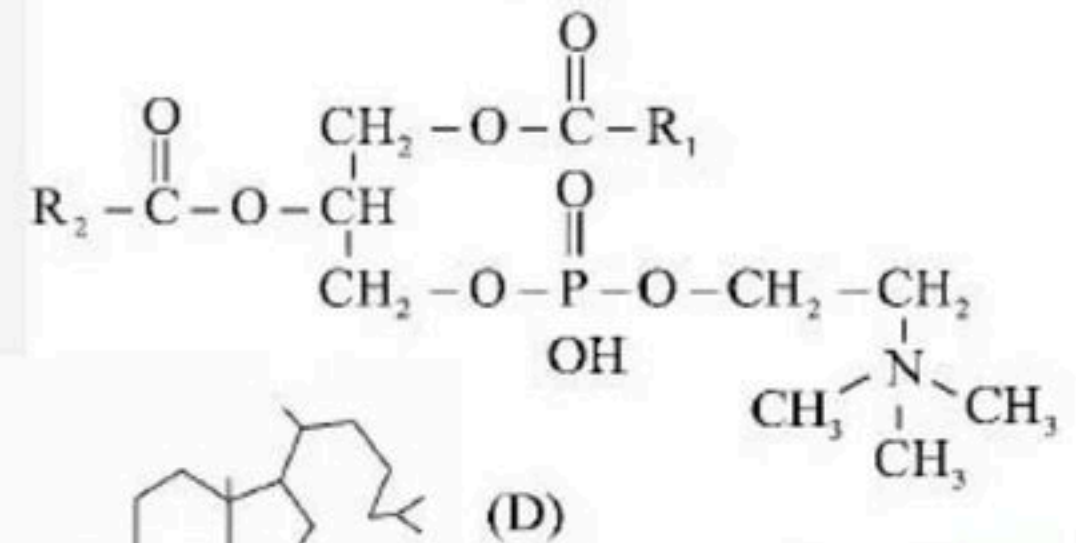
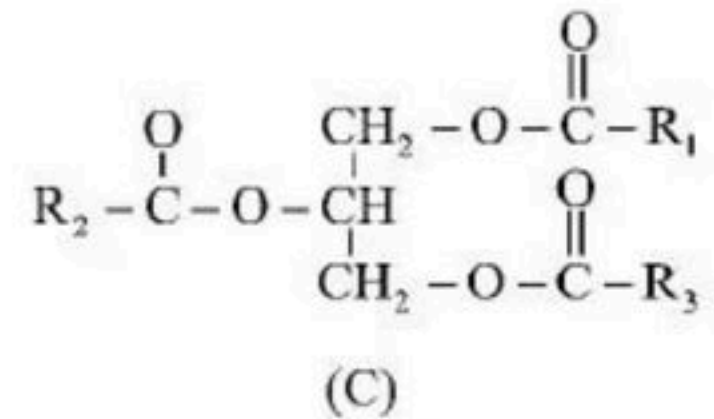
(3) C

(4) D



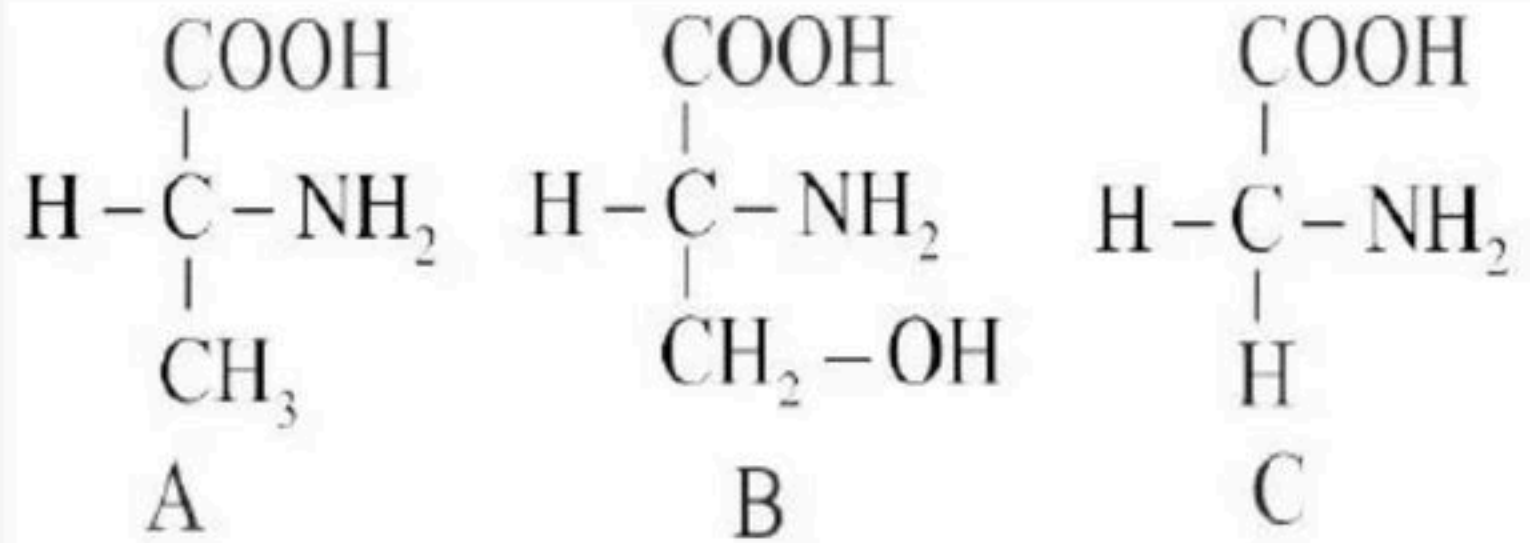
122. Identify A to E in the given diagram?

- (1) A - Glycerol, B - Fatty acid, C - Cholesterol,
D - Triglyceride, E - Fatty acid
- (2) A - Triglyceride, B - Glycerol, C - Fatty acid,
D - Lecithin, E - Cholesterol
- (3) A - Lecithin, B - Glycerol, C - Fatty acid,
D - Triglyceride, E -
Cholesterol
- (4) A - Fatty acid, B - Glycerol, C - Triglyceride,
D - Lecithin, E - Cholesterol



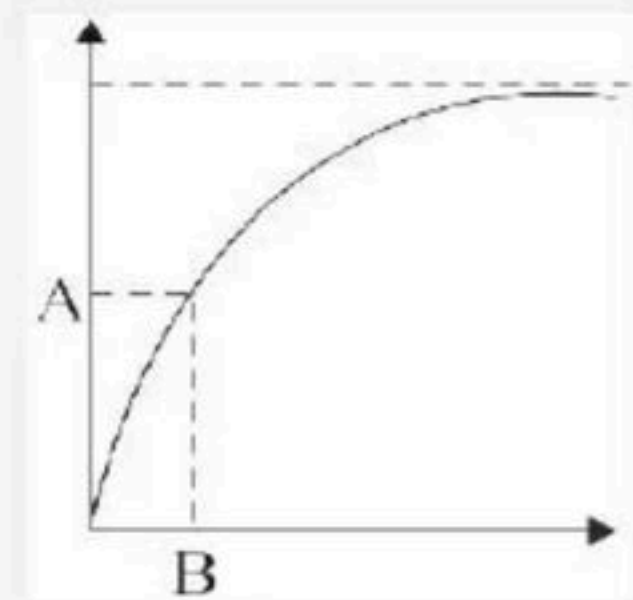
123. Identify A, B and C in the given diagram?

- (1) A - Alanine, B - Serine, C - Glycine
- (2) A - Glycine, B - Alanine, C - Serine
- (3) A - Glycine, B - Serine, C - Alanine
- (4) A - Serine, B - Glycine, C - Alanine



124. Identify A and B in the given diagram?

- (1) A - Velocity of reaction, B - Concentration of substrate
- (2) A - Velocity of reaction, B - Temperature
- (3) A - Velocity of reaction, B - K_m
- (4) A - Temperature, B - Velocity of reaction



125. Match the columns I (Protein) with II (Functions) and choose the correct option.

Column I

A Collagen

B Trypsin

C Insulin

D Antibody

(1) A-Q, B-P, C-S, D-R

(3) A-S, B-R, C-P, D-Q

Column II

P Enzyme

Q Intercellular ground substance

R Fights infectious agents

S Hormone

(2) A-P, B-S, C-Q, D-R

(4) A-R, B-S, C-P, D-Q



126. Select the incorrect statement from the following statements_

- (1) Primary metabolites have identifiable functions in host.
- (2) Many primary as well as secondary metabolites are useful for humans.
- (3) We know that all secondary metabolites in host play known roles in normal physiological process.
- (4) Some secondary metabolites have ecological importance.



127. Select the correct statement:

- (1) Increase in substrate concentration, increases the velocity of enzymatic reaction at first then it reaches maximum and further increment does not occur.
- (2) Increase in substrate concentration, decreases the velocity of enzymatic reaction at first then it reaches maximum and further increment does not occur.
- (3) Increase in substrate concentration, increases the velocity of enzymatic reaction at first then it reaches maximum and further decrease does not occur
- (4) All of these



128. Assertion : Cellulose is a homopolymer.

Reason : It consists of glucose as a monomer

- (1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.
- (2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- (3) Assertion is true but Reason is false.
- (4) Both Assertion and Reason are false.



129. Which of the following sugars have the same number of carbon as present in glucose?

- (1) Fructose
- (2) Erythrose
- (3) Ribulose
- (4) Ribose



130. In which of the following energy is released?

- (1) Conversion of glucose into pyruvate
- (2) Conversion of glucose into lactic acid
- (3) 1 and 2
- (4) Both are wrong



131. Which of the following is mismatched?

- (1) Chitin - polymer of glucosamine
- (2) Glycogen - Polymer of glucose
- (3) Cellulose - Heteropolysaccharide
- (4) Inulin - Homopolysaccharide



132. Match the columns I and II and choose the correct option.

Column I

A Pigment

B Alkaloid

C Terpenoid

D Essential oil

Column II

P Lemon grass oil

Q Diterpenes

R Anthocyanins

S Codeine

(1) A-S, B-R, C-Q, D-P

(3) A-R, B-S, C-Q, D-P

(2) A-S, B-R, C-P, D-Q

(4) A-R, B-P, C-Q, D-S



133. Match Column – I and Column – II with respect to proteins and their functions and select the correct option.

Column I

A Collagen

B Trypsin

C Insulin

D Antibody

Column II

P Fights infectious agents

Q Hormone

R Enzyme

S Intercellular ground substance

- (1) A – S, B – Q, C – R, D – P
- (2) A – S, B – R, C – Q, D – P
- (3) A – S, B – P, C – R, D – P
- (4) A – Q, B – R, C – Q, D – S



134. Match the columns I and II and choose the correct option.

Column I

A Nucleic acids catalyst

B Active site

C Proteins catalyst

D Inorganic catalysts

(1) A-S, B-R, C-P, D-Q

(2) A-R, B-S, C-Q, D-P

(3) A-R, B-S, C-P, D-Q

(4) A-Q, B-S, C-R, D-P

Column II

P Enzyme

Q Work efficiently at high temperatures and high pressures

R Ribozyme,

S Substrate binding site



135. Match the columns I and II and choose the correct option.

Column I

A Chitin

B N-acetyl galactosamine

C Cellulose

D Glycogen

Column II

P Plant cell walls

Q Chemically modified sugars

R Exoskeletons of arthropods

S Liver

(1) A-R, B-P, C-Q, D-S

(3) A-Q, B-R, C-S, D-P

(2) A-R, B-Q, C-P, D-S

(4) A-Q, B-P, C-S, D-R



136. Match Column-I with Column-II

Choose the correct answer from the options given below :

Column I**A GLUT-4****B Insulin****C Trypsin****D Collagen****Column II****P Hormone****Q Enzyme****R Intercellular ground substance****S Enables glucose transport into cell**

(1) A – S, B – P, C – Q, D – R

(2) A – P, B – Q, C – R, D – S

(3) A – Q, B – R, C – S, D – P

(4) A – R, B – S, C – P, D – Q



137. Match the columns I and II and choose the correct option.

Column I

A Glutamic acid

B Lysine

C Valine

D Tryptophan

(1) A-P, B-Q, C-S, D-R

(3) A-P, B-Q, C-R, D-S

Column II

P Basic amino acid

Q Acidic amino acid

R Aromatic amino acid

S Neutral amino acid

(2) A-Q, B-P, C-S, D-R

(4) A-P, B-R, C-S, D-Q



138. Match the columns I and II and choose the correct option.

Column I

A Oxidoreductases

B Transferases

C Hydrolases

D Isomerases

E Ligases

Column II

P Hydrolysis of bonds

Q Oxidoreduction between two substrates

R Linking together of two compounds

S Transfer of a group

T Inter-conversion of optical, geometric or positional isomers

(1) A-Q, B-P, C-T, D-S, E-R

(3) A-Q, B-S, C-P, D-T, E-R

(2) A-T, B-Q, C-P, D-R, E-S

(4) A-T, B-S, C-Q, D-R, E-P



139. Match the columns I and II and choose the correct option.

Column I

A Heteropolymer of amino acids

B Peptide bond

C Source of essential amino acids

D Collagen

E RuBisCO

(1) A-Q, B-R, C-P, D-S, E-T

(3) A-Q, B-P, C-R, D-T, E-S

Column II

P Formed between amino acids

Q Protein

R Dietary proteins

S Abundant protein in whole of the biosphere

T Most abundant protein in animal world

(2) A-R, B-Q, C-T, D-P, E-S

(4) A-P, B-Q, C-R, D-S, E-T



140. Statement A : The activity of an enzyme can be affected by a change in the conditions which alter the tertiary structure of the protein.

Statement B : These include temperature, pH, change in substrate concentration or binding of specific chemicals that regulate its activity.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



141. Consider the following statement :

(1) Coenzyme or metal ion that is tightly bound to enzyme protein is called prosthetic group.

(2) A complete catalytic active enzyme with its bound prosthetic group is called apoenzyme.

Select the correct option.

(1) Both (1) and (2) are true

(2) (1) is true but (2) is false.

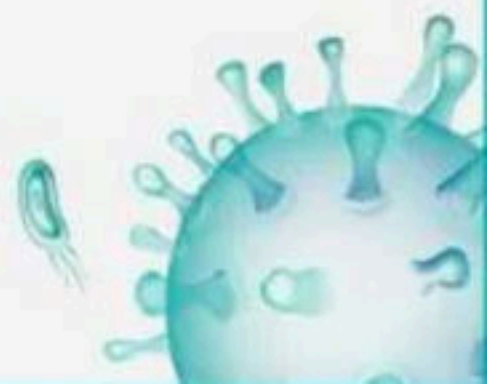
(3) Both (1) and (2) are false.

(4) (1) is false but (2) is true.



142. Statement A : Starch can hold I_2 molecules in the helical portion. Statement B : Starch- I_2 complex gives red colour.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



143. Statement A : Non-protein constituents, which are bound to the enzyme and make the enzyme catalytically active, are called cofactors.

Statement B : The protein portion of the enzymes is called the Apoenzyme.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



144. Statement A : Some proteins are an assembly of more than one polypeptide or subunits and it is called the quaternary structure.

Statement B : Adult human hemoglobin consists of 6 subunits or polypeptides.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



145. Statement A : Enzyme succinic dehydrogenase is inhibited by malonate by Non-competitive inhibition.

Statement B : Competitive inhibitors are often used in the control of bacterial pathogens.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



146. Statement A : Crossing over is the exchange of genetic material between two non-homologous chromosomes.

Statement B : Crossing over is an enzyme-mediated process and the enzyme involved is called recombinase.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



147. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Spindle fibres attach to kinetochores of chromosomes	P	Anaphase
B	Nucleolus, golgi complex and ER reform	Q	Telophase
C	Centromeres split and chromatids separate	R	Metaphase
D	Centrosome move towards opposite poles of the cell	S	Prophase

(1) A-P, B-R, C-S, D-Q

(2) A-R, B-Q, C-P, D-S

(3) A-P, B-S, C-Q, D-S

(4) A-P, B-R, C-Q, D-S



148. Match the columns I and II and choose the correct option.

Column I

A Nuclear membrane disappears by the end of

B Chromosomes align at the equator

C Meiosis ends with

D Splitting of the centromere of each chromosome

Column II

P Anaphase II

Q Telophase II

R Prophase II

S Metaphase-II

(1) A-P, B-S, C-R, D-Q

(2) A-R, B-S, C-Q, D-P

(3) A-Q, B-P, C-S, D-R

(4) A-Q, B-R, C-P, D-S



149. Match the columns I and II and choose the correct option.

Column I		Column II	
A	zygotene	P	Chromosomes become gradually visible under the light microscope
B	Leptotene	Q	Chromosomes start pairing together
C	Diplotene	R	Four chromatids of each bivalent chromosomes becomes distinct and clearly appears as tetrads
D	Pachytene	S	Dissolution of the synaptonemal complex

(1) A-Q, B-S, C-R, D-P

(2) A-S, B-P, C-Q, D-R

(3) A-Q, B-P, C-S, D-R

(4) A-P, B-Q, C-R, D-S



1. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Some lower plants and some social insects haploid cells divide by	P	Nucleocytoplasmic ratio.
B	Significant contribution of mitosis	Q	Mitosis
C	Essential for the cell to divide to restore	R	Continuously dividing cells
D	Cells of the upper layer of the epidermis	S	Cell repair

(1) A-P, B-Q, C-R, D-S

(2) A-Q, B-S, C-P, D-R

(3) A-S, B-R, C-Q, D-P

(4) A-S, B-R, C-P, D-Q



151. Match the columns I and II and choose the correct option.

Column I

A Synaptonemal complex formation

B Appears as tetrads

C Chromosomes become gradually visible under the light microscope

D X-shaped structures

Column II

P Pachytene

Q Leptotene

R Zygotene

S Chiasmata

(1) A-R, B-P, C-Q, D-S

(2) A-Q, B-P, C-R, D-S

(3) A-S, B-P, C-R, D-Q

(4) A-P, B-Q, C-R, D-S



152. Statement A : The sequence of events by which a cell duplicates its genome, synthesizes the other constituents of the cell and eventually divides into two daughter cells is termed cell cycle

Statement B : Cell division has to take place in a coordinated way to ensure correct division and formation of progeny cells containing intact genomes.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



153. Which does not occurs in prophase:

- (1) Formation of chromatin
- (2) Movement of centrioles
- (3) Appearance of sister chromatids
- (4) Compaction of chromosome



154. Statement A : The beginning of diplotene is recognised by the dissolution of the synaptonemal complex

Statement B : In oocytes of all vertebrates, diplotene can last for months or years.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



155. The two daughter cells formed during mitosis contains _____ as compared to the parent cell

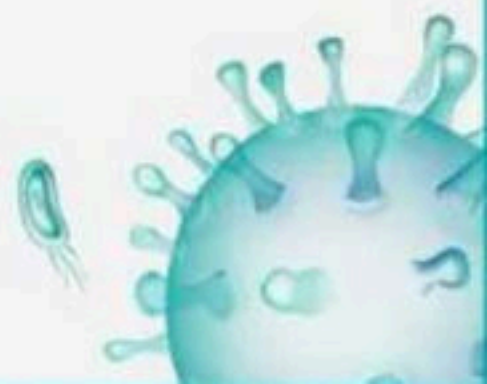
- (1) The same amount of DNA but half no. of chromosomes
- (2) The same amount of DNA and the same number of chromosomes
- (3) Half the amount of DNA and the half no. of chromosomes
- (4) Double the amount of DNA and double no. of chromosomes



156. Statement A : In some organisms cytokinesis is not followed by karyokinesis as a result of which multinucleate condition.

Statement B : Liquid endosperm in coconut is an example of Syncytium.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



157. Statement A : M-phase is the most dramatic period of the cell cycle, involving a major reorganization of virtually all components of the cell.

Statement B : In M-phase the number of chromosomes in the parent and progeny cells is the same, it is also called as equational division.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



158. Arrange the following events of meiosis in correct sequence :

I. Crossing over

II. Synapsis

III. Terminalisation of chiasmata

IV. Disappearance of nucleus

(1) II, III, IV, I

(2) II, I, IV, III

(3) II, I, III, IV

(4) I, II, III, IV



159. Statement A : At metaphase chromosome is made up of two sister chromatids, which are held together by the centromere.

Statement B : Small disc-shaped structures at the surface of the centromeres are called Centrosome.

- (1) Statement-A is correct but Statement-B is incorrect.
- (2) Statement-A is incorrect but Statement-B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



160. Given below are two statements:

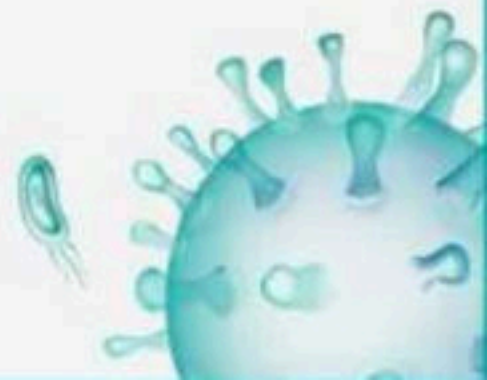
Statement A : During G_0 phase of cell cycle, the cell is metabolically inactive.

Statement B : The centrosome undergoes duplication during S phase of interphase.

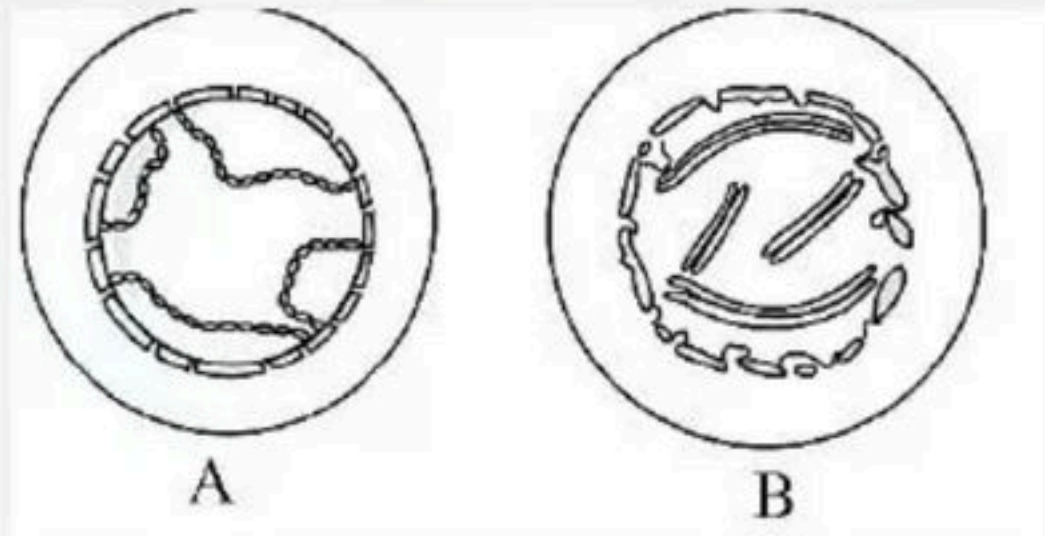
In the light of the above statements, choose the most appropriate answer from the

options given below:

- (1) Both Statement A and Statement B are correct
- (2) Both Statement A and Statement B are incorrect.
- (3) Statement A is correct but Statement B is incorrect
- (4) Statement A is incorrect but Statement B is correct.



161. In the given diagram A and B represents_



- | | |
|------------------------|-------------------|
| (1) A - Early Prophase | B - Metaphase |
| (2) A - Early Prophase | B - Late Prophase |
| (3) A - Early Anaphase | B - Late Anaphase |
| (4) A - Anaphase | B - Metaphase |



162. Match the following Columns I & II and choose the correct option.

Column I

A CO_2 is added to RUBP

B CO_2 is added to PEP

C Photosynthesis

D Photorespiration

(1) A-P , B-R , C-Q , D-S

(3) A-P , B-R , C-S , D-Q

Column II

P OAA (4C)

Q High O_2 , Low CO_2

R Phosphoglycerate (3C)

S Redox process

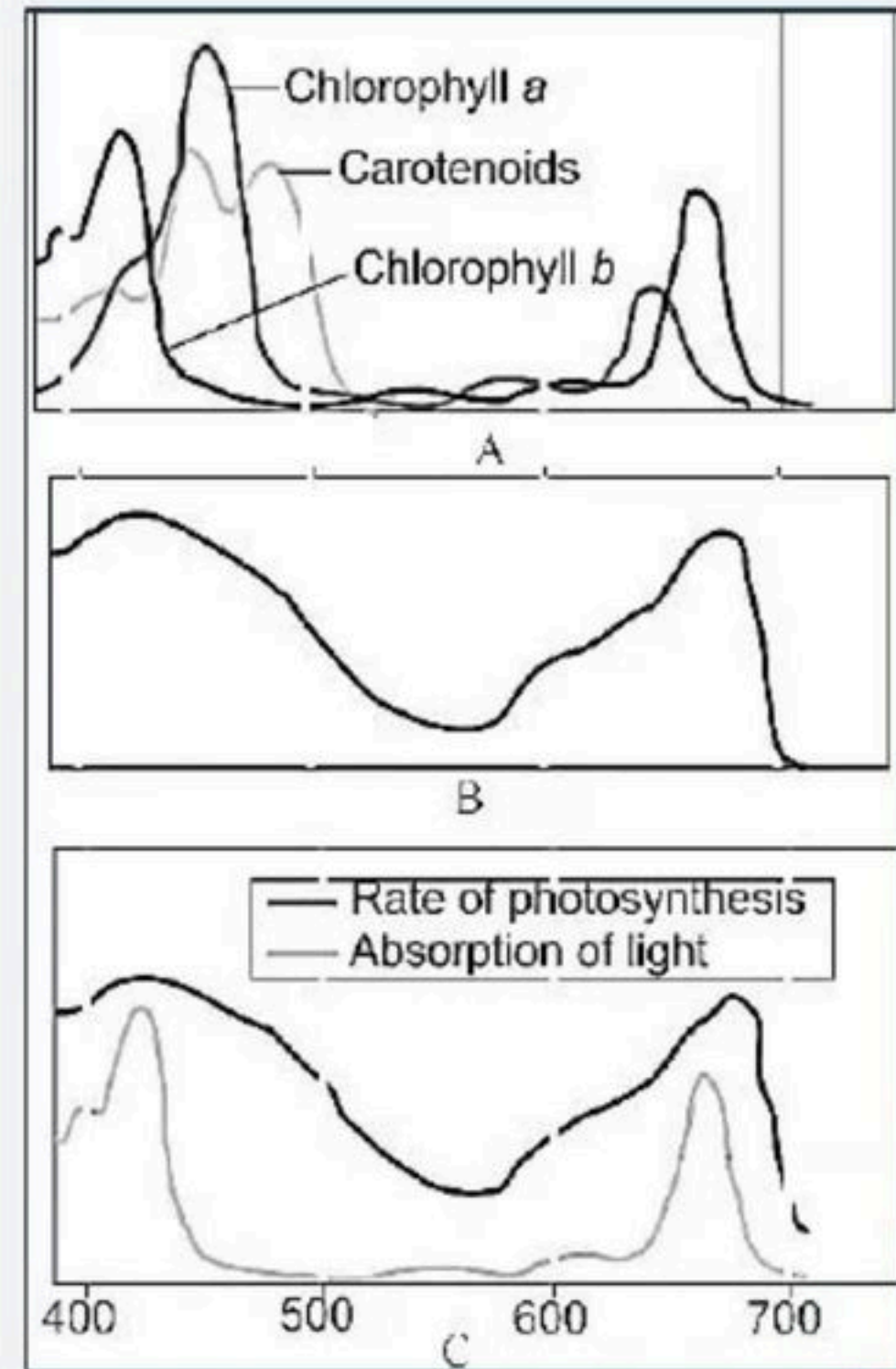
(2) A-R , B-P , C-Q , D-S

(4) A-R , B-P , C-S , D-Q



163. Following graph shows action spectrum and absorption spectrum. Identify in which regions of the spectrum, most of the photosynthesis take place?

- (1) Blue light only
- (2) Red light only
- (3) Both 1 and 2
- (4) None of the above



164. In plants some O_2 does bind to RuBisCO, as result of this__

- (1) O_2 fixation is decreased
- (2) CO_2 fixation is decreased
- (3) CO_2 fixation is increased
- (4) Both (1) and (3)



165. The initial enzyme of Calvin cycle is

- (1) Ribulose 1,5 diphosphate carboxylase
- (2) Ribulose 1,3 diphosphate carboxylase
- (3) Hexokinase
- (4) Cytochrome oxidase



166. An experiment using two leaves, a variegated leaf or a leaf that was covered with black paper and one that was exposed to light on testing these leaves for starch shows that photosynthesis occurs only in _____ part of leaf and in the _____ of sunlight

- (1) Non-green, presence
- (2) Green, presence
- (3) Non-green, absence
- (4) Green, absence



167. Read the following statements regarding photorespiration and select the total number of correct statements.

- (a) In C_3 plants some O_2 does bind to RuBisCO, and hence CO_2 fixation is increased.
- (b) In the photorespiratory pathway, there is neither synthesis of sugars, nor of ATP.
- (c) Photorespiration results in the release of O_2 with the utilisation of ATP.
- (d) In the photorespiratory pathway there is no synthesis of ATP or NADPH.
- (e) The biological function of photorespiration is not known yet.

(1) 4

(2) 2

(3) 5

(4) 3



168. Given below are some characteristics of photosynthetic plants. Which of these are true for C_4 plants

- i) RuBisCo is the primary and first acceptor of CO_2**
- ii) The first stable compound formed is 4-carbon compound**
- iii) The first stable compound formed is PGA**
- iv) Possess PEP carboxylase**
- v) The first carboxylation reaction occurs in bundle sheath cells**
- vi) The HSK cycle takes place in mesophyll cells**
- vii) 3-carbon in the primary CO_2 acceptor**

- | | |
|------------------------------------|-----------------------------------|
| (1) (i), (iv), (v), (vii) | (2) (i), (ii), (iii), (iv), (vii) |
| (3) (ii), (iii), (iv), (vi), (vii) | (4) (ii), (iv), (vi), (vii) |



169. Match the following Columns I & II and choose the correct option.

Column I		Column II	
A	Non-cyclic Photophosphorylation	P	Only ATP is formed
B	Chemiosmosis	Q	Water splitting complex
C	Cyclic Photophosphorylation	R	Lumen of Thylakoid
D	Photosystem II	S	Z-scheme

(1) A-Q , B-P , C-S , D-R

(2) A-Q , B-S , C-P , D-R

(3) A-P , B-Q , C-R , D-S

(4) A-S , B-R , C-P , D-Q



170. Match the pigments given in Column I(Pigments) with their colour in chromatogram given in Column II(Colour in chromatogram).

Choose the correct option from the following :

Column I		Column II	
A	Chlorophyll 'b'	P	Yellow orange
B	Carotenoids	Q	Orange red
C	Chlorophyll 'a'	R	Yellow
D	Xanthophylls	S	Blue green
		T	Yellow green

(1) A – S, B – T, C – R, D – Q

(2) A – P, B – Q, C – R, D – T

(3) A – T, B – P, C – S, D – R

(4) A – T, B – P, C – R, D – S



171. Match the columns I & II and choose the correct option.

Column I

A Complex I

B Complex II

C Complex III

D Complex IV

Column II

P Succinate dehydrogenase

Q NADH dehydrogenase

R Cytochrome c oxidase

S Cytochrome bc₁ complex

(1) A-Q, B-P, C-R, D-S

(2) A-R, B-Q, C-S, D-P

(3) A-Q, B-P, C-S, D-R

(4) A-P, B-Q, C-R, D-S



172. Match Column I with Column II :

Choose the correct answer from the options given below :

Column I		Column II	
A	Oxidative decarboxylation	P	Citrate synthase
B	Glycolysis	Q	Pyruvate dehydrogenase
C	Oxidative phosphorylation	R	Electron transport system
D	Tricarboxylic acid cycle	S	EMP pathway

(1) A – R, B – S, C – Q, D – P

(2) A – Q, B – S, C – P, D – R

(3) A – R, B – P, C – Q, D – S

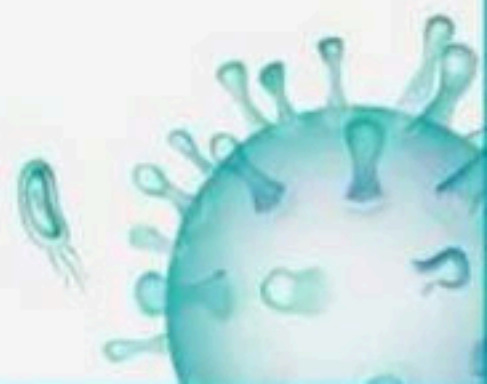
(4) A – Q, B – S, C – R, D – P



173. Statement A : Pyruvic acid is transported into the mitochondria where it is converted into acetyl CoA with the release of O_2

Statement B : Acetyl CoA enters the Krebs' cycle operating in the matrix of mitochondria

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



174. Statement A : When carbohydrates are used as substrate and are completely oxidised, the RQ will be 1

Statement B : Equal amounts of CO_2 and O_2 are evolved and consumed, when carbohydrates are used as substrate

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



175. Statement A : Under anaerobic conditions either lactic acid fermentation or alcohol fermentation occurs

Statement B : Fermentation takes place under anaerobic conditions in many prokaryotes and multicellular eukaryotes

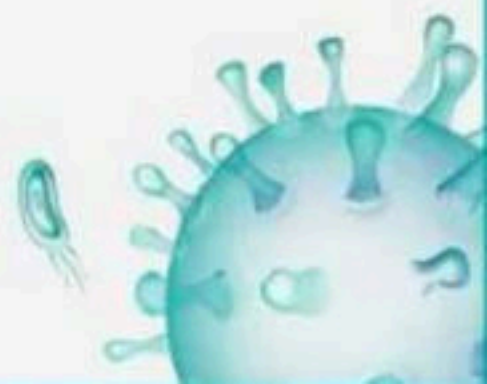
- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



176. Statement A : Auxin help to prevent fruit and leaf drop at early stages.

Statement B : Auxin promotes the abscission of older mature leaves and fruits.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



177. Match the columns I and II.

Column I

A Indole compounds

B Adenine derivatives

C Derivatives of carotenoids

D Terpenes

Column II

P IAA

Q Absciscic acid, ABA

R N⁶-furfurylamino purine, kinetin

S Gibberellic acid, GA₃

(1) A – R, B – P, C – Q, D – S

(3) A – P, B – R, C – Q, D – S

(2) A – S, B – R, C – P, D – Q

(4) A – Q, B – P, C – R, D – S



178. Match the entries in Column I with those of Column II and choose the correct answer.

Column I**A Cytokinin****B Auxins****C Abscissic acid****D . Ethylene****Column II****P Stress hormone****Q Ripening of fruits****R Apical dominance****S Bolting****T .Richmond Lang effect****(1) A - T, B - R, C - P, D - Q****(3) A - R, B - S, C - Q, D - P****(2) A - T, B - R, C - Q, D - S****(4) A - Q, B - Q, C -T,D - R**

179. Match the columns I and II.

Column I		Column II	
A	Auxin	P	Hastening the maturity period of conifers
B	Cytokinins	Q	Xylem differentiation
C	Gibberelins	R	Overcome the apical dominance
D	Ethylene	S	Tolerance of plants to various kinds of stresses
E	Abscissic acid	T	Breaks seed and bud dormancy

(1) A – R, B – P, C – Q, D – T, E – S

(2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q

(4) A – Q, B – T, C – P, D – R, E – S



180. Match the columns I and II.

Column I		Column II	
A	Growth at a cellular level	P	17,500 new cells per hour
B	Maize root apical mersitem	Q	Increase in size by upto 3,50,000 times
C	Watermelon cells	R	Increase in the amount of protoplasm
D	Growth of a pollen tube	S	Measurement in terms of surface area
E	Growth in a dorsiventral leaf	T	Measurement in terms of length

(1) A – R, B – P, C – Q, D – T, E – S

(2) A – Q, B – R, C – P, D – T, E – S

(3) A – S, B – R, C – P, D – T, E – Q

(4) A – Q, B – T, C – P, D – R, E – S



181. Statement A : In the alveoli, there is high pO_2 , low pCO_2 , lesser H^+ concentration and higher temperature, the factors are all favourable for the formation of oxyhaemoglobin

Statement B : In the tissues, where low pO_2 , high pCO_2 , high H^+ concentration and higher temperature exist, it results in dissociation of oxygen from the oxyhaemoglobin.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



182. Statement A : The lungs are situated in the thoracic chamber which is anatomically an air-tight chamber.

Statement B : The thoracic chamber is formed ventrally by the vertebral column, dorsally by the sternum, laterally by the ribs and on the lower side by the dome-shaped diaphragm.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



183. Match the items given in column I with those in column II and select the correct option given below:

Column I**A Tidal volume****B Inspiratory reserve volume****C Expiratory reserve volume****D Residual volume****Column II****P 2500-3000 MI****Q 1100-1200 MI****R 500-550 mL****S 1000-1100 MI**

(1) A – P, B – S, C – Q, D – R

(2) A – R, B – P, C – S, D – Q

(3) A – R, B – Q, C – P, D – S

(4) A – S, B – R, C – Q, D – P



184. Match the column:

Column I		Column II	
A	Asthma	P	Inflammation of bronchi and bronchioles
B	Emphysema	Q	Chronic disorder in which alveolar walls are damaged due to which respiratory surface is decreased.
C	Occupational respiratory disorders	R	Long exposure of chemicals can give rise to inflammation leading to fibrosis (proliferation of fibrous tissues).

- (1) A – P, B – Q, C – R
(2) A – Q, B – P, C – R
(3) A – R, B – P, C – Q
(4) A – R, B – Q, C – P



185. Match Column I with Column II :

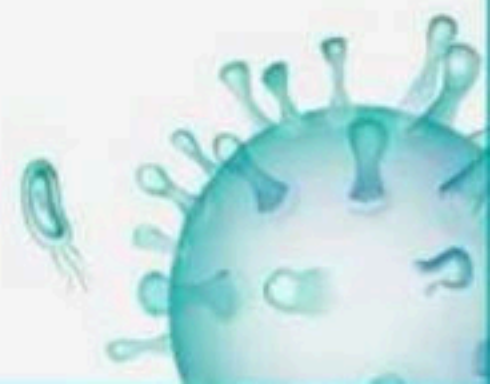
Column I		Column II	
A	Expiratory capacity	P	Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
B	Functional residual capacity	Q	Tidal volume + Expiratory reserve volume
C	Vital capacity	R	Tidal volume + Inspiratory reserve volume
D	Inspiratory capacity	S	Expiratory reserve volume + Residual volume

(1) A – Q, B – S, C – P, D – R

(2) A – R, B – Q, C – S, D – P

(3) A – Q, B – P, C – S, D – R

(4) A – P, B – R, C – Q, D – S



186. Match the following and mark the correct options (animal respiratory organ).

Column I

A Earthworm

B Terrestrial arthropods

C Fishes

D Birds/Reptiles

Column II

P Moist skin

Q Gills

R Lungs

S Trachea

- (1) A – Q, B – P, C – S, D – R
- (2) A – P, B – S, C – Q, D – R
- (3) A – P, B – R, C – Q, D – S
- (4) A – P, B – R, C – S, D – R



187. Assertion : Human lungs have enormous surface area for the exchange of gases.

Reason : In human, glottis is guarded by epiglottis.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



188. Assertion : Larynx is a cartilaginous box which help in sound production.

Reason : Epiglottis flap covers the glottis during swallowing and prevents the entry of food into the larynx.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



189. Assertion : A chemosensitive area is situated adjacent to the rhythm centre which is highly sensitive to and hydrogen ions.

Reason : The role of oxygen in the regulated of respiratory rhythm is quite insignificant.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



190. Conducting part of the respiratory system is formed

- (1) From external nostrils up to the initial bronchioles
- (2) From external nostrils up to the duct of alveoli
- (3) From external nostrils up to the terminal bronchioles
- (4) From external nostrils up to the alveoli



191. Which of the following events do not occur during joint diastole?

i. Blood from the pulmonary veins and vena cava flows into the left and right ventricles respectively through the left and right atria

ii. The semilunar valves are closed.

iii. Action potential is conducted from SAN to AVN.

iv. All four chambers of heart are in relaxed state.

v. Tricuspid and bicuspid valves open.

(1) Only i, iv

(2) Only iii

(3) Only i, ii

(4) Only i, ii and v



192. Mitral valve is found between

- (1) Left atrium and left ventricle
- (2) Right atrium and right ventricle
- (3) Right atrium and left ventricle
- (4) Left atrium and right ventricle



193. Which of the following statement is incorrect?

- (1) The body has the ability to alter the stroke volume as well as the heart rate and thereby the cardiac output.
- (2) In each minute a single cardiac cycle is performed.
- (3) During each cardiac cycle two prominent sounds are produced.
- (4) Cardiac cycle includes atrial systole, ventricular systole and joint diastole.



194. How many people are ve in human population?

- (1) 80%
- (2) 50%
- (3) 60%
- (4) 100%



195. Mark the incorrect statement regarding normal ECG.

- (1) Patient is connected to 3 electrical leads (one to each wrist and to the left ankle).
- (2) T' wave represents atrial repolarization.
- (3) Q marks the beginning of ventricular depolarization.
- (4) R represents ventricular depolarization.



196. How many statements are wrong?

- i. Systemic circulation takes place between left ventricle and right atrium of heart.**
- ii. Adrenal medullary hormones can increase the cardiac output.**
- iii. Heart failure is not the same as cardiac arrest or heart attack**
- iv. The opening between the right atrium and the right ventricle is guarded by a valve formed by two muscular flops**

(1) 1

(2) 2

(3) 3

(4) 4



197. Statement A : Heart is protected by a double walled membranous bag, pleural membrane enclosing the pericardial fluid.

Statement B : Mammalian heart has four chambers, two relatively small upper chambers called atria and two larger lower chambers called ventricles.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



198. Identify the correct statement

I. Arteries are blood vessels that carry blood away from the heart

II. Aorta is the main artery leaving the heart

III. Pulmonary artery is the only artery that carries oxygen

(1) I and III

(2) I and II

(3) II and III

(4) All are correct



199. Match the columns I and II and choose the correct option.

Column I

A Heart beat

B Cardiac cycle

C Stroke volume

D Cardiac output

Column II

P 5 L

Q 70 ml

R 0.8 sec

S 72 per minute

(1) A-P, B-R, C-Q, D-S

(2) A-Q, B-P, C-R, D-S

(3) A-Q, B-R, C-S, D-P

(4) A-S, B-R, C-Q, D-P



200. Find the incorrect matching:

- (1) CAD-Atherosclerosis
- (2) Pain in chest-Angina pectoris
- (3) Hypertension-BP raised
- (4) Heart failure-Heart attack



THANK YOU



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